

Appendices

Habitat heterogeneity and phenotypic variation: Site temperature largely predicts diverse spawning portfolios of an interior Chinook Salmon stock

27 February, 2026

Contents

A1	Datasets	2
A1.1	Spawn timing data	2
A1.2	Covariate Derivation and Screening	7
A2	Exploratory Data Analysis	10
A2.1	Bivariate relationships	10
A2.2	Stream temperature (temp_90)	12
A2.3	mean_elevation	15
A2.4	slope	17
A2.5	flow_90	18
A3	Data Analysis	21
A3.1	Initial Model Selection	21
A3.2	Targeted model comparison	23
A3.3	Residual Diagnostics	24
A3.4	Evaluation of Temperature Functional Form	25
A3.5	Interactions	27
A4	Final Model Diagnostics and Interpretation	28
A4.1	Residual diagnostics	28
A4.2	Model parameters and performance	31
A4.3	Predictions against original data	34
A4.4	Population-level effects	35
A4.5	Group-level effects (deviations from fixed effects)	39

A1 Datasets

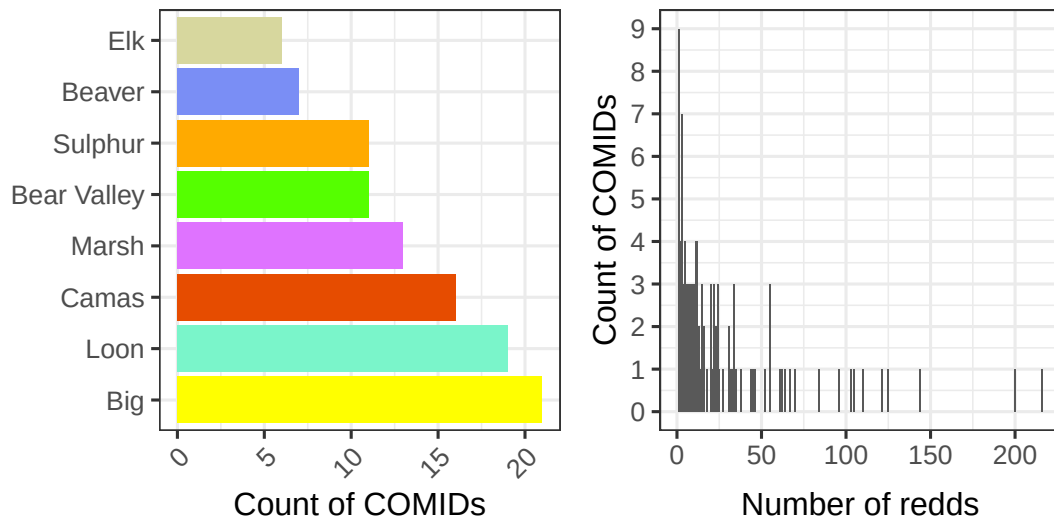
A1.1 Spawn timing data

Spawn timing data for Chinook salmon were collected from 2001 to 2005 in the MFSR. Because redds were not observed daily, we inferred spawn dates as the initial date (day of year) a completed redd was observed. We removed data from 2001, and data from Knapp Creek and Cape Horn Creek, as these sites were not consistently sampled. The final dataset contained 3016 georeferenced redds (i.e., inferred spawn date) across 104 stream reaches (i.e., COMIDs), nested within six major spawning tributaries that also included Beaver Creek (tributary to Marsh Creek) and Elk Creek (tributary to Bear Valley Creek), totaling eight spawning streams across four years.

We spatially joined each redd GPS location to the NHDPlus Version 2 (Horizon Systems, 2018) to assign stream reaches based on a common identifier (COMID). The COMID is used to link redd data with covariate data associated with the stream reach on which it is located.

Grouping Structure

The structure of the data is repeated measures on COMIDs, and multiple years, shared across COMIDs within streams. Many COMIDs only have 1-2 observation (Figure A1.1). There are 23 COMIDs with <5 redds (26%), 13 with ≤ 2 . (12.5%).



Distribution of spawn timing

Spawn timing is generally unimodal, with a peak in late August to early September (Figure A1.2A). The distribution is slightly left-skewed, but the mean and median are similar, indicating low skewness, and the variance is low with no evidence of overdispersion. There is also substantial variation in spawn date by stream and year (Fig. A1.2B-C; Fig. A1.3) and within COMIDs within streams (Fig. A1.4).

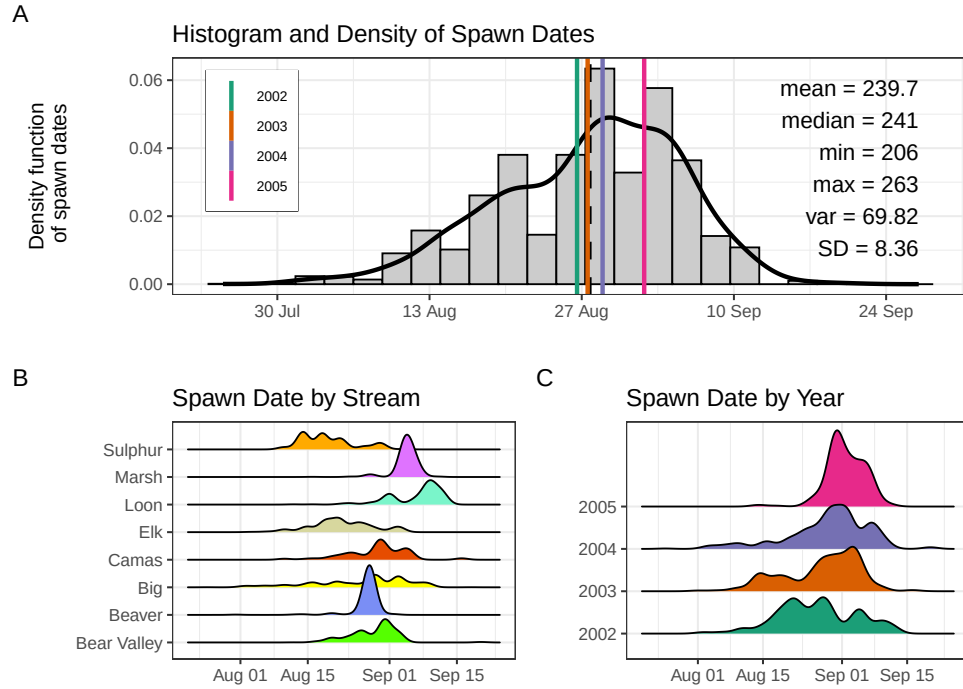


Figure A1.2: Histogram and density of spawn timing for all streams and years (A), and by stream (B) and year (C). In panel (A), colored, vertical lines indicate the mean spawn date for each year., and the black vertical line indicates the global mean.

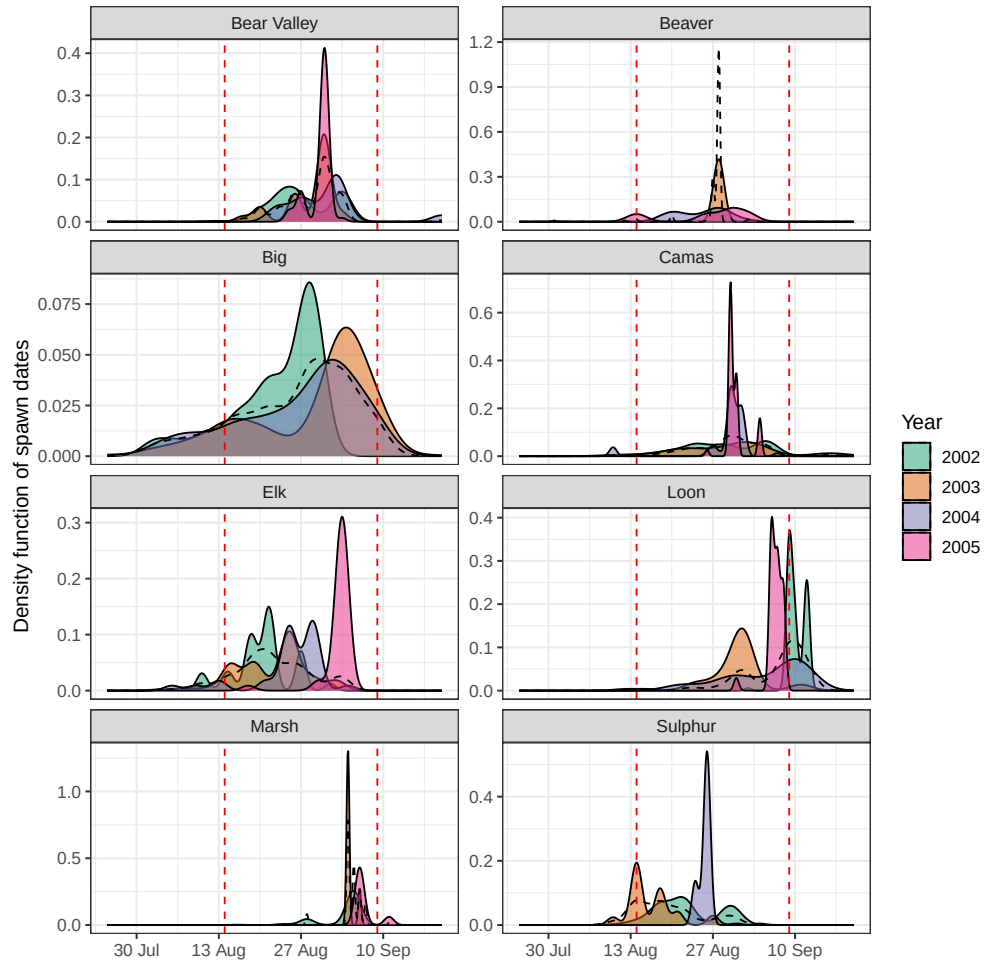


Figure A1.3: Kernel density plots showing the overall distribution of spawn dates by stream and year.

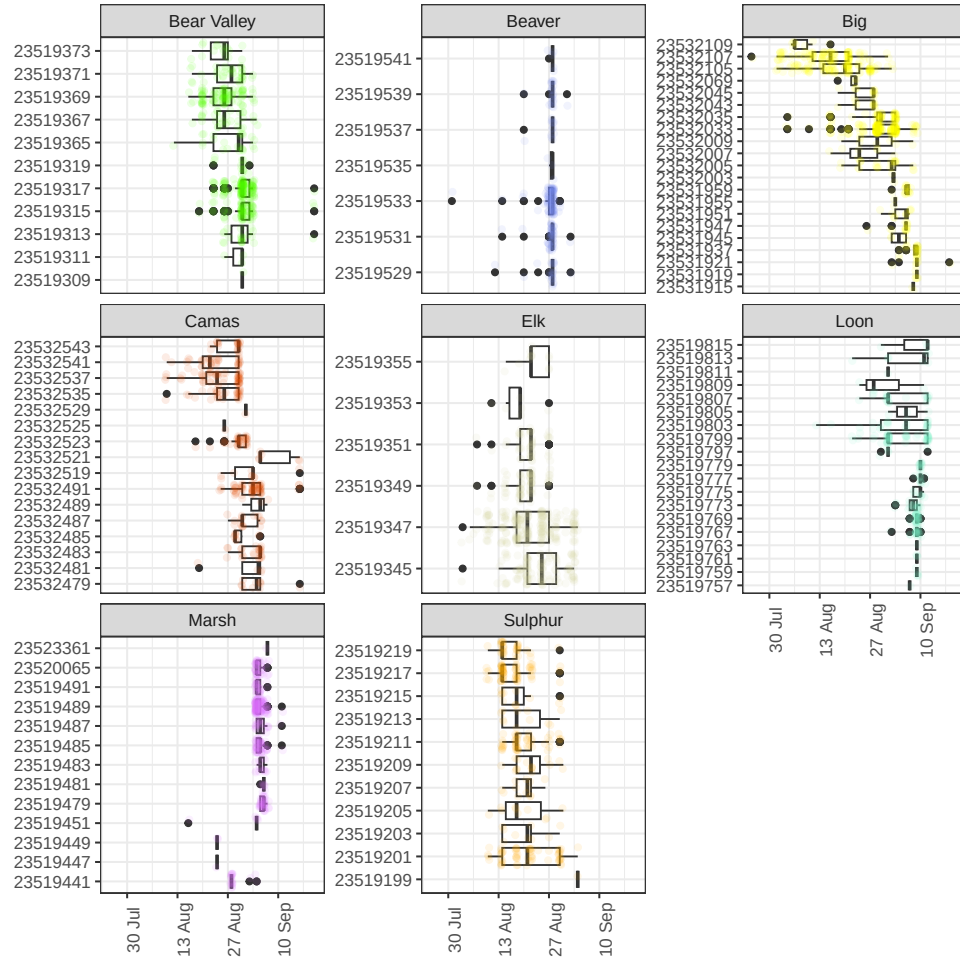


Figure A1.4: Variation in Chinooks salmon spawn timing by COMID within streams.

Cumulative proportion of redds over time

The temporal progression of Chinook salmon spawning activity varied within each stream across the four study years (Figure A1.5). The cumulative proportion of redds provides an intuitive measure of the spawning season's pace and timing. Streams like Marsh and Sulphur exhibit rapid increases, suggesting short, concentrated spawning windows. In contrast, streams like Big and Camas show more gradual increases, indicating a more protracted spawning season. Year-to-year variation is evident in several streams. Overall, this figure highlights both spatial and temporal heterogeneity in spawn timing that underpins the need for models incorporating stream- and year-specific effects.

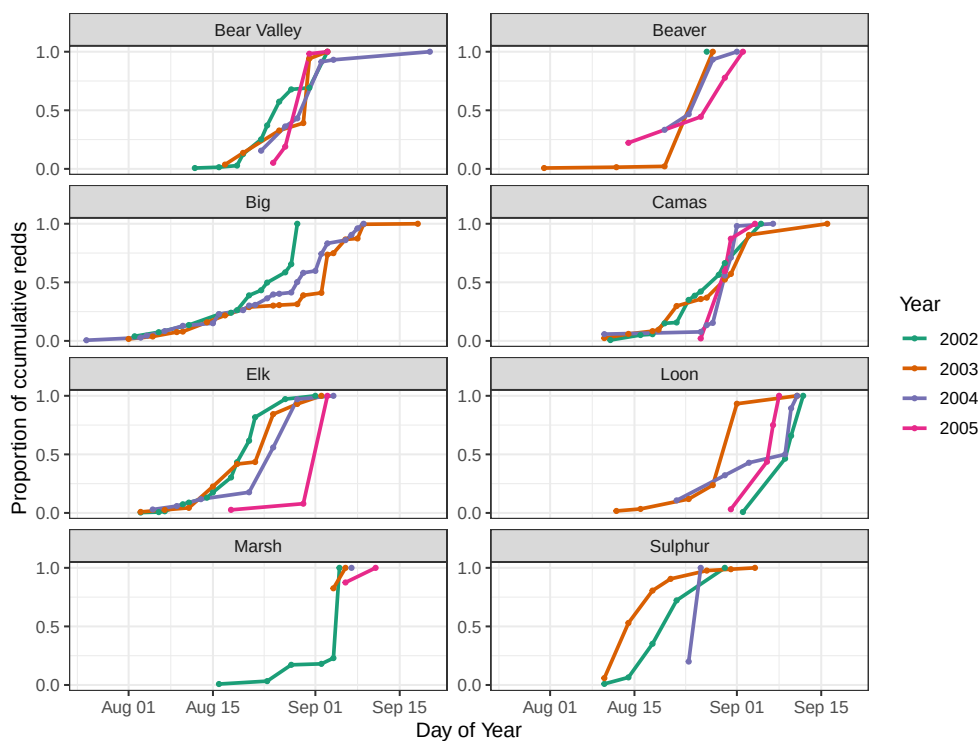


Figure A1.5: Proportion of cumulative Chinook salmon redds over time (day of year) across years (2002–2005) and streams. Each line represents a different year, with color denoting the year. Stream-specific panels illustrate temporal variation in the progression of spawning activity, as measured by cumulative redd counts normalized to the maximum value in each stream-year combination.

A1.2 Covariate Derivation and Screening

To identify environmental predictors of Chinook salmon spawn timing, we quantified covariates that describe thermal and physical stream conditions at redd locations. Covariates were selected based on three criteria: (1) demonstrated influence on salmon phenology in prior literature, (2) availability at all redd locations (i.e., COMIDs), and (3) low collinearity with other predictors.

The initial focal covariates included stream temperature (°C), stream discharge (cms), elevation (m), and stream slope (unitless). Elevation and slope were extracted from NHDPlus (Moore et al. 2019).

Elevation and slope

Elevation and stream slope data were available at the COMID (stream reach) scale from the NHDPlus Version 2 (Horizon Systems, 2018).

Stream temperature

We used modeled daily average stream temperature values predicted at the stream segment (COMID) scale (Siegel et al. 2023). These data were downloaded and filtered to 2002-2005 and for the MFSR. Figure A1.6 shows the modeled thermal regimes for MFSR tributaries.

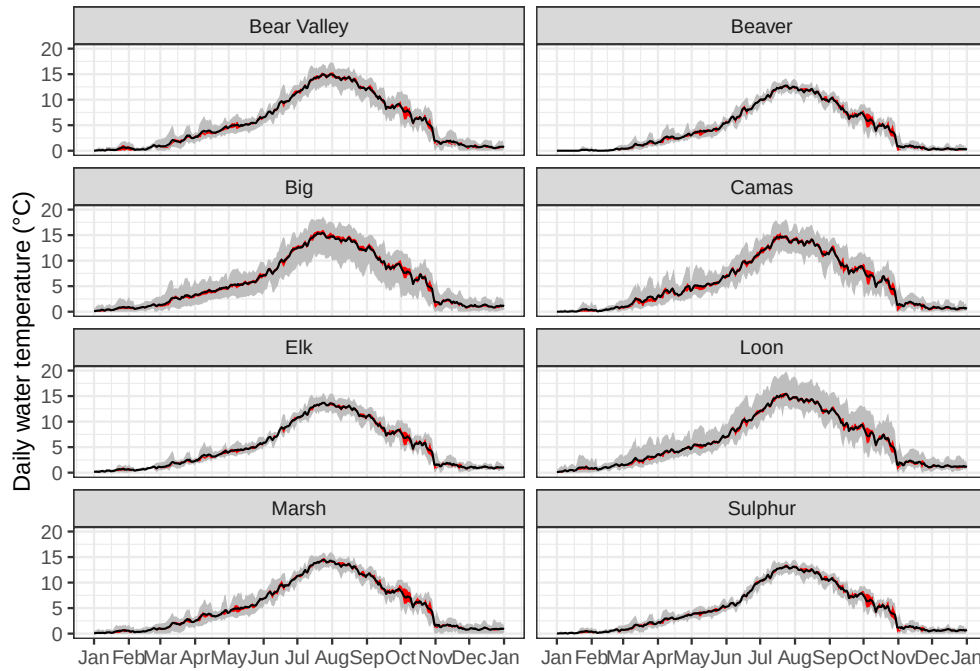


Figure A1.6: Modeled thermal regimes (2001-2005) for MFSR tributaries. Black line are means, Red ribbon are 40 to 60th percentiles, Grey ribbon are full range.

These modeled temperatures were validated against empirical logger data collected at a subset of sites in the Middle Fork basin, with strong agreement (Pearson correlation coefficient = 0.97; Figure A1.7).

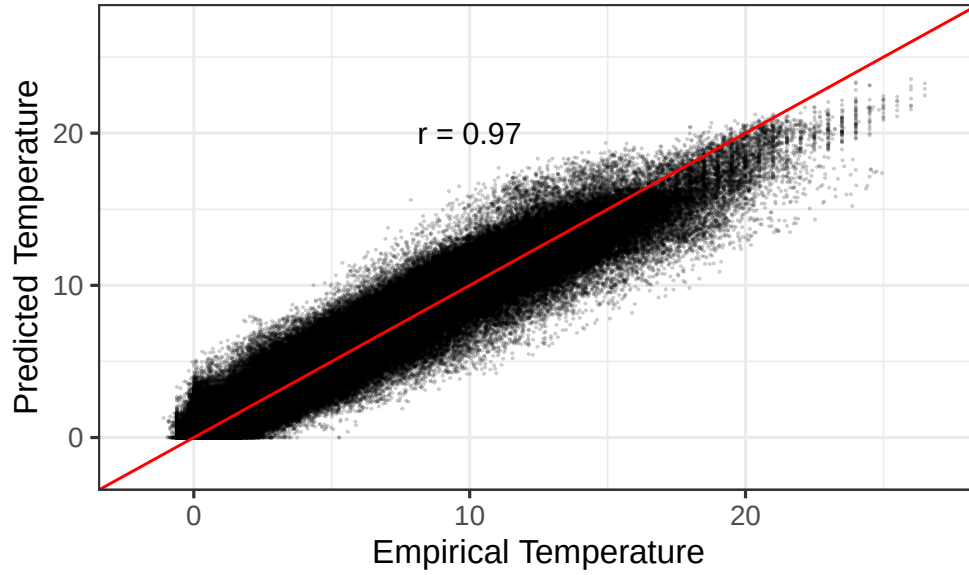


Figure A1.7: Comparison of modeled stream temperatures (Siegel et al. 2023) against empirical logger data for the Middle Fork Salmon River.

Discharge (streamflow)

Stream flow data were compiled from a single USGS Gage lower in the watershed (MF Salmon River at MF Lodge NR Yellow Pine ID - 13309220; Figure A1.8).

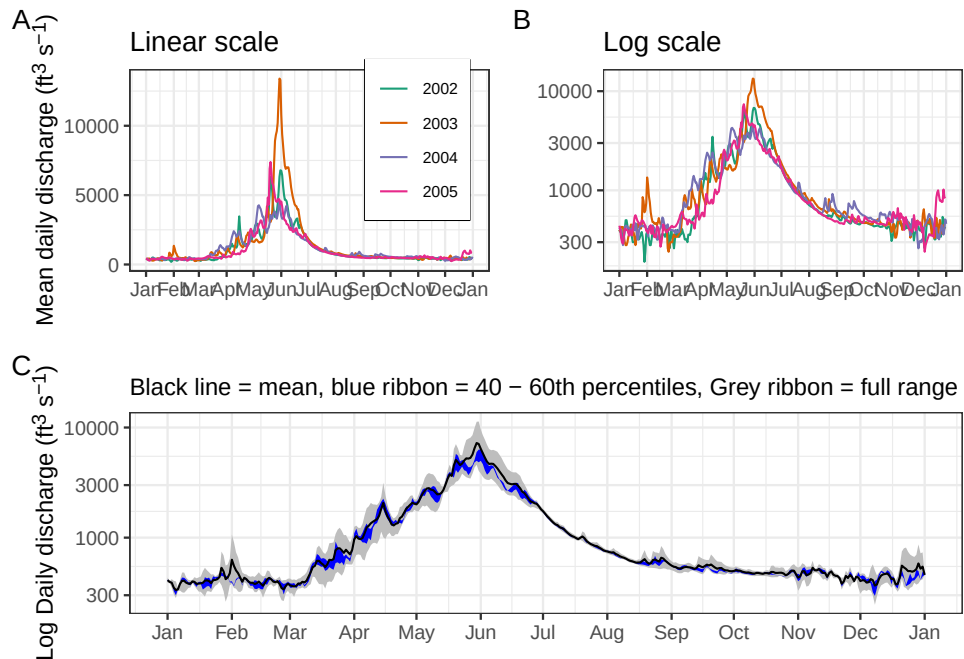


Figure A1.8: Inter-annual variability in daily discharge (cfs) at MF Lodge USGS Gage 13309220.

Time-windowed temp and flow metrics

For each redd, we computed several time-windowed summaries of temperature and flow relative to the inferred spawn date:

- *Antecedent metrics*: average conditions over 30, 60, and 90 days preceding the spawn date
- *Spanning metrics*: averages over time windows centered on the spawn date,
- *Post-spawn metrics*: 30-, 60-, and 90-day averages following the spawn date,
- *Time-invariant metrics*: summaries calculated relative to a fixed calendar date (e.g., August 1) representing an approximate onset of the spawning window.

The time invariant and after metrics were omitted from further consideration as preliminary data exploration showed weak if any relationship with spawn timing.

A2 Exploratory Data Analysis

A2.1 Bivariate relationships

We conducted exploratory data analysis on the compiled dataset to identify candidate covariates for model selection. Bivariate relationships between spawn date (yday) and continuous predictors (Figure A2.1), and pairwise correlations between all continuous covariates (Figure A2.2), are shown below.

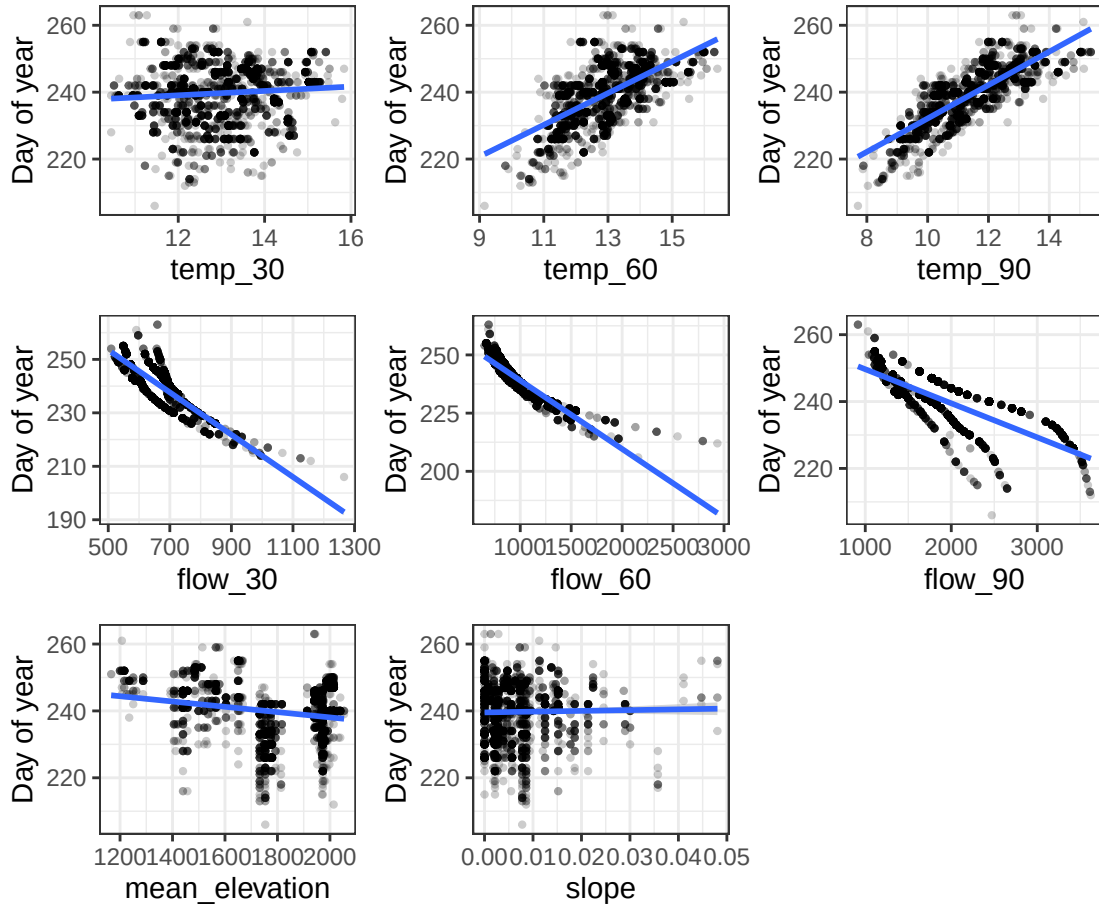


Figure A2.1: Bivariate relationships between spawn date ('yday') and continuous covariates. Solid lines are linear fits.

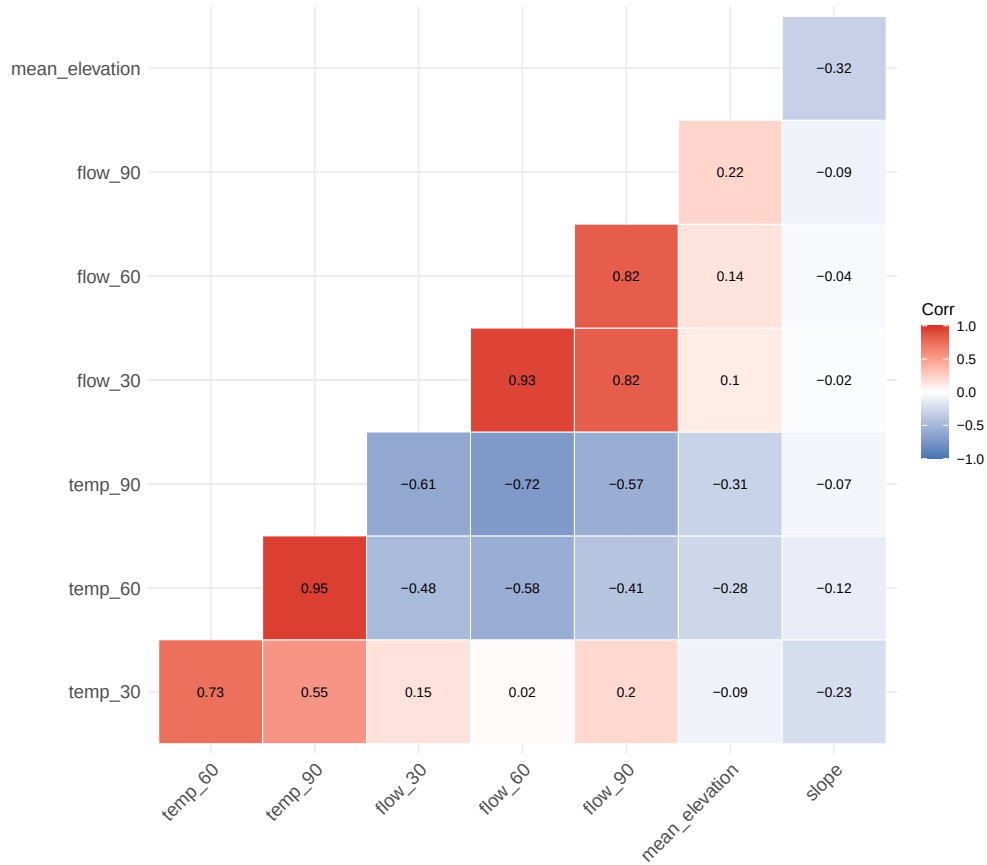


Figure A2.2: Pairwise correlations between continuous covariates.

Takeaways from Figures A2.1 and A2.2:

Among temperature variables, **temp_90** clearly shows the strongest relationship with spawn date (Figure A2.1) and is highly collinear with **temp_30** and **temp_60** (Figure A2.2), supporting its use as the sole temperature predictor. There is a weak negative relationship between **mean_elevation** and **yday**, and it is only weakly correlated with **temp_90** ($r = 0.31$). Flow variables (**flow_30**, **flow_60**, **flow_90**) all show a decaying-exponential relationship with spawn date, with a strong year effect that suggests much of their explanatory signal is redundant with year as a fixed effect. Stream **slope** shows no meaningful relationship with spawn date.

A2.2 Stream temperature (temp_90)

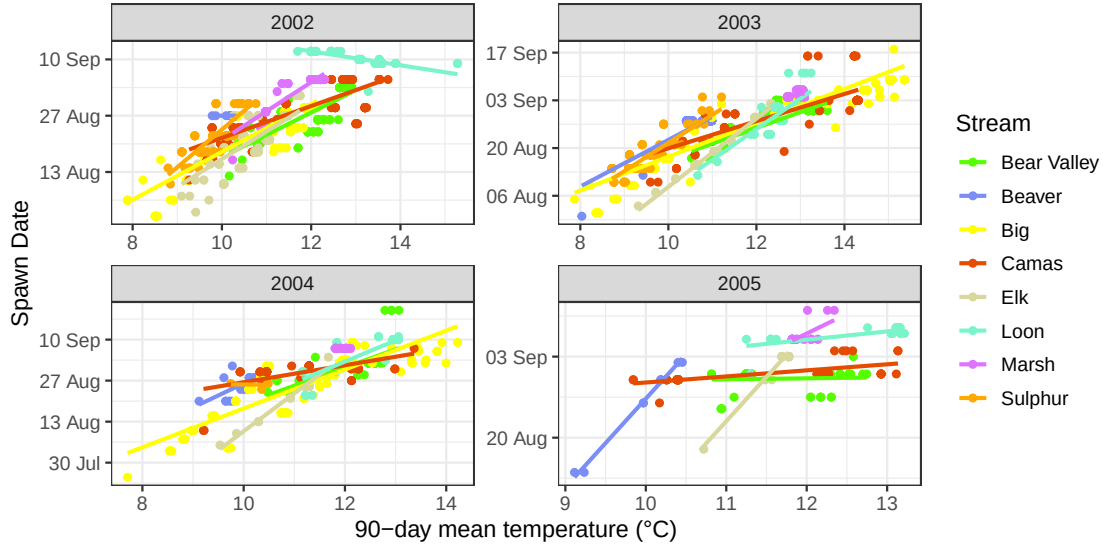


Figure A2.3: Relationship between 90 day antecedent stream temperature and spawn date by stream and year. Solid lines are linear fits.

Given observed positive relationships (Figure A2.3), we evaluated the role of `temp_90` in explaining variation in Chinook salmon spawn timing:

```
# temp_90 models
m.t90.1 <- lmer(yday ~ (1|COMID), data = df_mod, REML = FALSE)
m.t90.2 <- lmer(yday ~ temp_90 + (1|COMID), data = df_mod, REML = FALSE)
m.t90.3 <- lmer(yday ~ temp_90 + I(temp_90^2) + (1|COMID), data = df_mod, REML = FALSE)
```

The model with a quadratic effect of temperature (`m.t90.3`) improved dramatically over the null (intercept-only) model and linear effect (Table A2.1), and boosting both conditional and marginal R^2 (Table A2.2). This confirms a strong, nonlinear effect of pre-spawn temperature on spawn timing (Figure A2.4).

Table A2.1: AIC comparison of linear mixed models relating spawn date ('yday') to 'temp_90'.

Model	df	AIC	delta
m.t90.3	5	15848.81	0.0000
m.t90.2	4	16570.12	721.3107
m.t90.1	3	18629.30	2780.4987

Table A2.2: Model performance of linear mixed models relating spawn date ('yday') to 'temp_90'.

Name	R2_conditional	R2_marginal	ICC	RMSE	Performance_Score
m.t90.3	0.9177407	0.7643117	0.6509825	3.108315	0.8333333
m.t90.2	0.9086448	0.6915320	0.7038423	3.489863	0.6100606
m.t90.1	0.6571743	0.0000000	0.6571743	4.929494	0.0195227

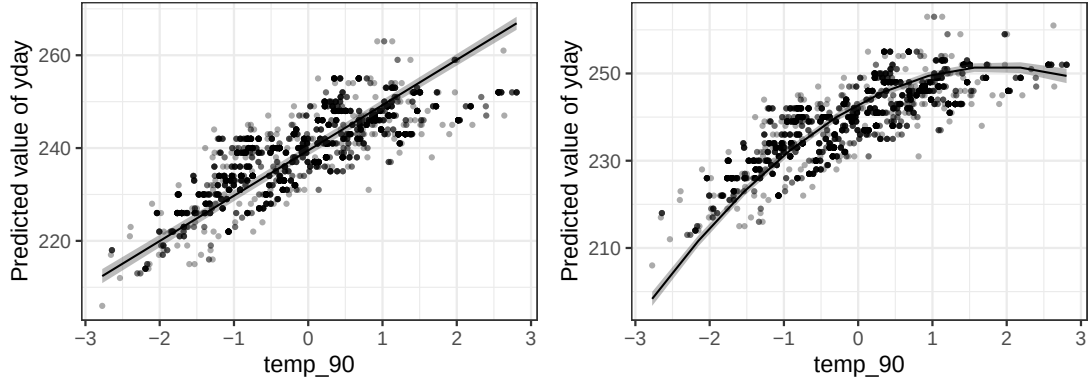


Figure A2.4: Marginal effect of 90 day antecedent stream temperature on spawn timing for (A) ‘m.t90.2’ and (B) ‘m.t90.3’.

Next, we add in stream and year as main effects, and then interactions:

```
# temp + stream + year models
m.t90.4 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + (1 | COMID), data = df_mod,
  REML = FALSE)
m.t90.5 <- lmer(yday ~ temp_90 + I(temp_90^2) + year + (1 | COMID), data = df_mod,
  REML = FALSE)
m.t90.6 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + year + (1 | COMID),
  data = df_mod, REML = FALSE)
m.t90.7 <- lmer(yday ~ temp_90 * stream + I(temp_90^2) + year + (1 | COMID),
  data = df_mod, REML = FALSE)
m.t90.8 <- lmer(yday ~ temp_90 * year + I(temp_90^2) + stream + (1 | COMID),
  data = df_mod, REML = FALSE)
```

Table A2.3: AIC comparison of linear mixed models relating spawn timing to temperature.

Model	df	AIC	delta
m.t90.8	18	12136.83	0.0000
m.t90.7	22	13038.01	901.1783
m.t90.6	15	13473.12	1336.2837
m.t90.5	8	13531.17	1394.3321
m.t90.4	12	15807.98	3671.1454
m.t90.3	5	15848.81	3711.9726

Table A2.4: Model performance of linear mixed models relating spawn timing to temperature.

Name	R2_conditional	R2_marginal	ICC	RMSE	Performance_Score
m.t90.8	0.9928189	0.6448836	0.9797782	1.581472	0.8333333
m.t90.7	0.9852284	0.7177821	0.9476590	1.865263	0.5267941
m.t90.6	0.9800601	0.7293886	0.9263154	2.022117	0.5061439
m.t90.5	0.9871140	0.6600787	0.9620912	2.021778	0.4528690
m.t90.3	0.9177407	0.7643117	0.6509825	3.108315	0.2196386
m.t90.4	0.8982595	0.7921275	0.5105629	3.110542	0.1666667

Adding stream (`m.t90.4`) or year (`m.t90.5`) individually improved model fit relative to `m.t90.3`, but year alone (`m.t90.5`) performed better than stream alone (`m.t90.4`) (Table A2.3).

Combining stream and year additively (`m.t90.6`) provided further improvement in AIC, along with the best marginal R^2 and RMSE (Table A2.4). This suggests that both stream and year provide meaningful structure in explaining variation in spawn timing. Adding the interaction between temperature and stream (`m.t90.7`) increased complexity, slightly lowered the marginal R^2 , and has high multicollinearity (VIF ~ 15).

Adding the interaction between temperature and year (`m.t90.8`) gave the lowest AIC overall but had the lowest marginal R^2 of all models tested. This means that while the model “fit” the data better on paper (AIC), it did so by overfitting or by capturing year-to-year idiosyncrasies that may not generalize well to new data (suspicious marginal effect, Figure A2.5).

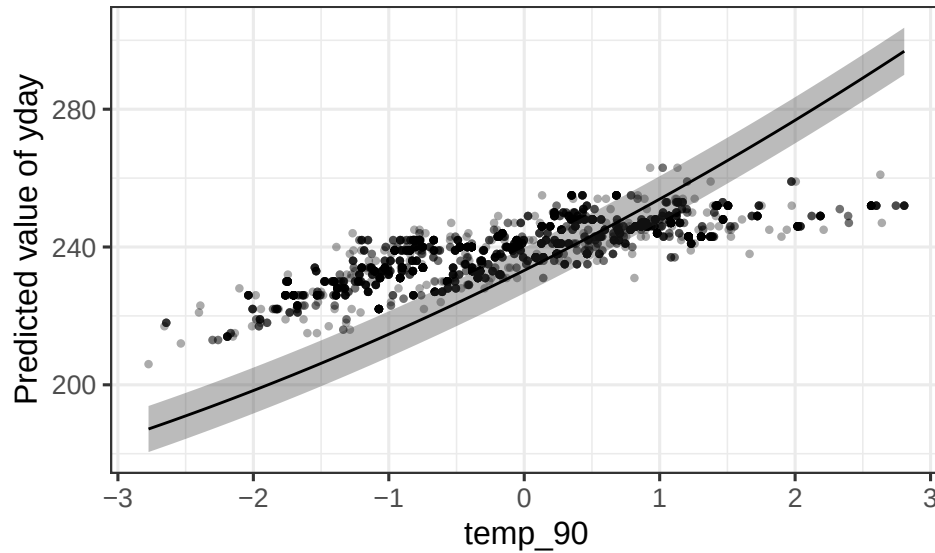


Figure A2.5: Marginal effect (`'m.t90.8'`) of 90 day antecedent stream temperature on spawn timing.

The suspicious curvature in the marginal effect plot of `m.t90.8` (showing an unrealistic predicted relationship between `temp_90` and `yday`) confirms the risk of confounding or overfitting with interaction terms—particularly given that the random intercepts already capture substantial site-level variation.

Given the trade-offs, model `m.t90.6` emerged as the best compromise: it retained additive main effects of temperature, stream, and year, had high conditional and marginal R^2 , moderate AIC, low RMSE, and avoided high multicollinearity.

A2.3 mean_elevation

Figure A2.6 shows the relationship between (A) `mean_elevation` and `temp_90` by stream and (B) `mean_elevation` and `yday` (spawn date) by stream and year. Solid lines are linear fits.

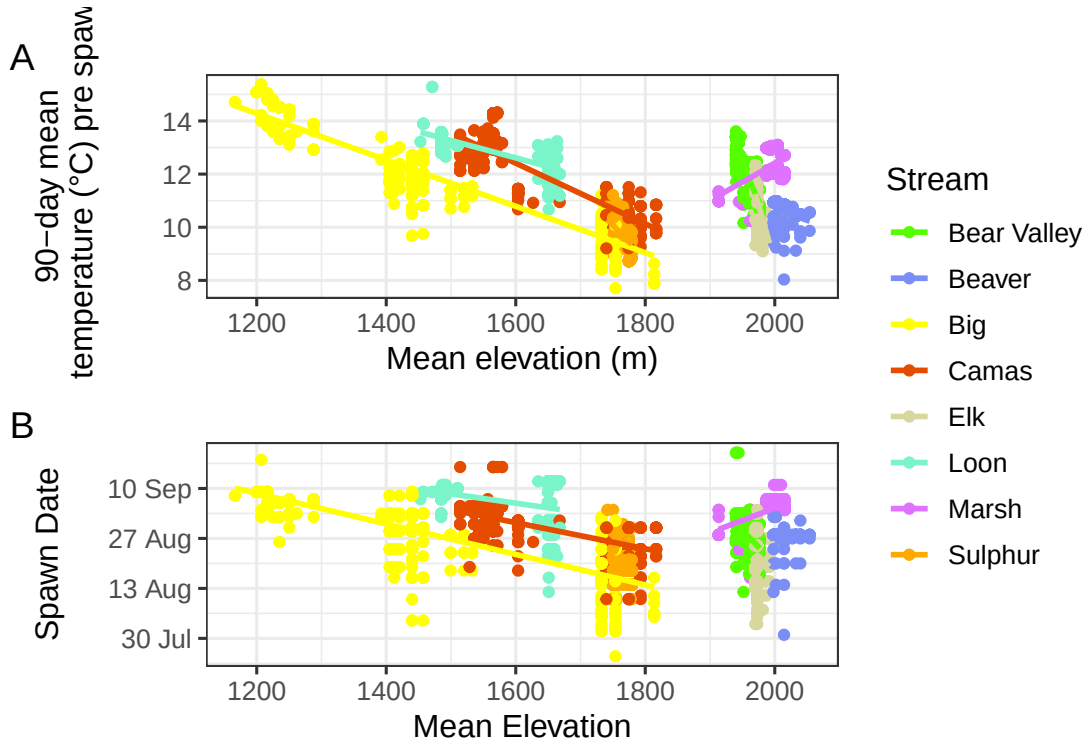


Figure A2.6: Relationship between (A) `mean_elevation` and `temp_90` by stream and (B) `mean_elevation` and spawn timing by stream and year. Solid lines are linear fits.

We evaluated the role of mean elevation in explaining variation in Chinook salmon spawn timing:

```
# mean elevation models
m.ele.1 <- lmer(yday ~ (1|COMID), data = df_mod, REML = FALSE)
m.ele.2 <- lmer(yday ~ mean_elevation + (1|COMID), data = df_mod, REML = FALSE)
m.ele.3 <- lmer(yday ~ mean_elevation + stream + (1|COMID), data = df_mod, REML = FALSE)
m.ele.4 <- lmer(yday ~ mean_elevation + year + (1|COMID), data = df_mod, REML = FALSE)
m.ele.5 <- lmer(yday ~ mean_elevation + stream + year + (1|COMID), data = df_mod, REML = FALSE)
m.ele.6 <- lmer(yday ~ mean_elevation * stream + (1|COMID), data = df_mod, REML = FALSE)
m.ele.7 <- lmer(yday ~ mean_elevation * year + (1|COMID), data = df_mod, REML = FALSE)
```

Including mean elevation as a main effect improved model fit ($\Delta\text{AIC} = 19$ between `m.ele.1` and `m.ele.2`). The best-supported model (`m.ele.5`) included mean elevation alongside stream and year ($\Delta\text{AIC} = 281$ between `m.ele.2` and `m.ele.5`), with moderate collinearity (VIFs ~ 6), suggesting that mean elevation can contribute to explaining spawn timing when controlling for stream and year.

However, adding interactions with stream or year substantially increased model complexity and collinearity, as evidenced by extremely high VIFs (>45). Given the known inverse relationship between elevation and temperature across streams (Figure A2.6A), caution is warranted when later incorporating both elevation and temperature in the same model.

Table A2.5: AIC comparison of linear mixed models relating spawn timing to mean_elevation.

Model	df	AIC	delta
m.ele.5	14	18329.29	0.00000
m.ele.6	18	18371.62	42.33148
m.ele.3	11	18431.88	102.58771
m.ele.7	10	18435.12	105.82567
m.ele.4	7	18501.44	172.14764
m.ele.2	4	18611.27	281.97710
m.ele.1	3	18629.30	300.01337

Table A2.6: Model performance of linear mixed models relating spawn timing to mean_elevation.

Name	R2_conditional	R2_marginal	ICC	RMSE	Sigma	AICc_wt	Performance_Score
m.ele.5	0.6399737	0.5939968	0.1132428	4.876512	4.927506	1	0.6843974
m.ele.7	0.6684326	0.1243275	0.6213568	4.774852	4.851946	0	0.5175646
m.ele.4	0.6574860	0.1155940	0.6127186	4.835441	4.913296	0	0.4006749
m.ele.1	0.6571743	0.0000000	0.6571743	4.929494	5.009972	0	0.2596896
m.ele.2	0.6503692	0.1014398	0.6108988	4.929830	5.009158	0	0.2505245
m.ele.6	0.6398941	0.6301923	0.0262347	4.989054	5.016809	0	0.1678027
m.ele.3	0.6269782	0.5838335	0.1036716	4.971609	5.022058	0	0.1413263

Table A2.7: Variance inflation factors for model m.ele.5.

	GVIF	Df	$\text{GVIF}^{1/(2 \cdot \text{Df})}$
mean_elevation	5.739424	1	2.395709
stream	5.943076	7	1.135760
year	1.042077	3	1.006893

Table A2.8: Variance inflation factors for model m.ele.6.

	GVIF	Df	$\text{GVIF}^{1/(2 \cdot \text{Df})}$
mean_elevation	2.317261e+03	1	48.137937
stream	9.622172e+10	7	6.088629
mean_elevation:stream	5.226657e+11	7	6.870938

A2.4 slope

slope is not related to yday (Figure A2.7A), though there is some association between slope and yday when considering stream (Figure A2.7B). We will likely drop slope.

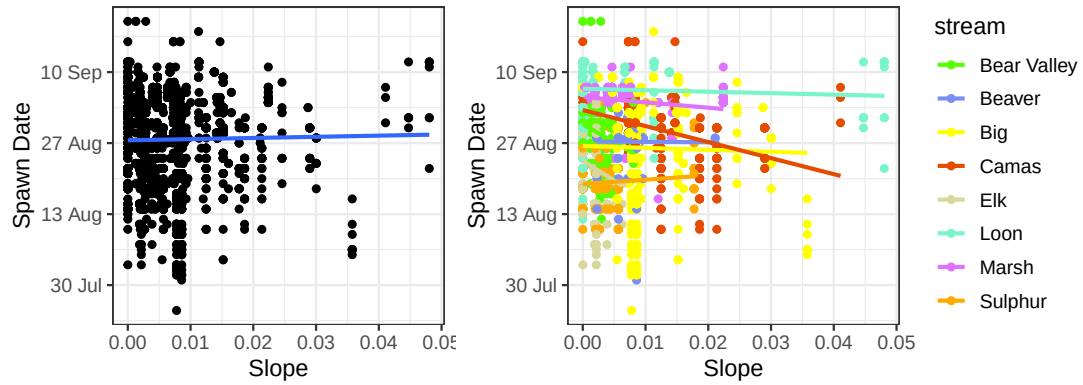


Figure A2.7: Bivariate relationship between 'slope' and spawn date (A) and by stream (B). Solid lines are linear fits.

A2.5 flow_90

Because flow data are not COMID- or stream-specific, it makes sense to think about and represent flow as an out-of-basin year effect that determines when adults make it back to the MFSR and initially onto the spawning grounds. The strong correlation between `flow_90` and `year` can be seen clearly in (Figure A2.8A).

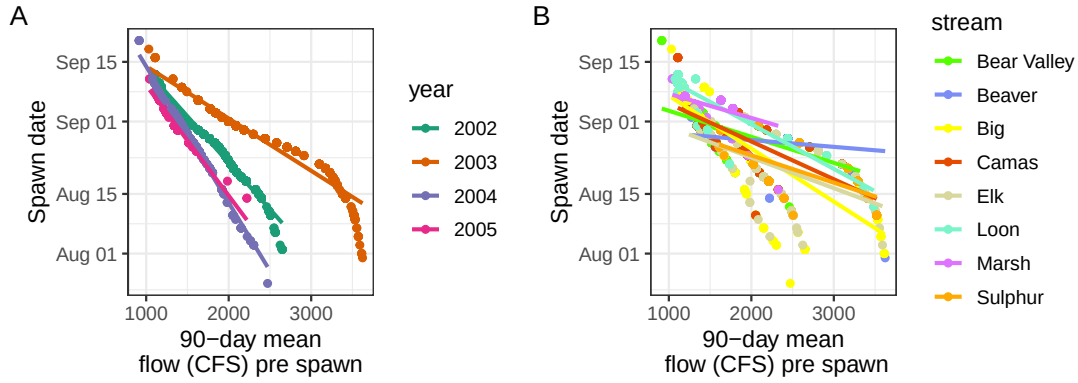


Figure A2.8: Spawn timing vs. 90-day mean discharge pre spawn at MF Lodge.

Next we compare the following simple linear models to examine functional structure:

```
# flow_90 models
m.flow.1 <- lm(yday ~ flow_90, data = model_data)
m.flow.2 <- lm(yday ~ I(flow_90^2), data = model_data)
m.flow.3 <- lm(yday ~ flow_90 + year, data = model_data)
m.flow.4 <- lm(yday ~ flow_90 + year + stream, data = model_data)
m.flow.5 <- lm(yday ~ flow_90 * stream, data = model_data)
m.flow.6 <- lm(yday ~ flow_90 * stream + year, data = model_data)
m.flow.7 <- lm(yday ~ flow_90 * year + stream, data = model_data)
m.flow.8 <- lm(yday ~ flow_90 * stream + I(flow_90^2), data = model_data)
m.flow.9 <- lm(yday ~ flow_90 * stream + year + I(flow_90^2), data = model_data)
```

Table A2.9: Model performance of linear models relating spawn date ('yday') to 'flow_90'.

Name	R2	R2_adjusted	RMSE	Sigma	AICc_wt	Performance_Score
m.flow.7	0.9780529	0.9779505	1.237710	1.240800	1	1.0000000
m.flow.9	0.9407030	0.9403269	2.034449	2.041229	0	0.4960534
m.flow.6	0.9108990	0.9103638	2.493858	2.501751	0	0.4472891
m.flow.4	0.8913935	0.8909958	2.753330	2.758824	0	0.4181935
m.flow.3	0.8745347	0.8743681	2.959322	2.961778	0	0.3942842
m.flow.8	0.7348812	0.7334668	4.301804	4.313980	0	0.2174139
m.flow.5	0.7130109	0.7115760	4.475722	4.487641	0	0.1921569
m.flow.1	0.5900974	0.5899614	5.348977	5.350752	0	0.0576227
m.flow.2	0.5352432	0.5350890	5.695650	5.697539	0	0.0000000

Table A2.10: Parameter estimates for m.flow.7

Parameter	Coefficient	SE	CI_low	CI_high	p
(Intercept)	278.2566486	0.2130585	277.8388932	278.6744041	0.0000000
flow_90	-0.0219390	0.0001136	-0.0221617	-0.0217163	0.0000000
year2003	-8.3607844	0.2371172	-8.8257131	-7.8958557	0.0000000
year2004	11.0240656	0.3856186	10.2679620	11.7801692	0.0000000
year2005	1.0123700	0.7619753	-0.4816766	2.5064167	0.1840768
streamBeaver	0.5591186	0.1048594	0.3535151	0.7647220	0.0000001
streamBig	-0.4395848	0.0772852	-0.5911220	-0.2880475	0.0000000
streamCamas	-0.0934454	0.0899470	-0.2698094	0.0829186	0.2989375
streamElk	0.2219142	0.0861007	0.0530918	0.3907365	0.0100026
streamLoon	0.0895360	0.1073905	-0.1210304	0.3001024	0.4044923
streamMarsh	-0.7143977	0.0913025	-0.8934196	-0.5353759	0.0000000
streamSulphur	-0.0665471	0.1017872	-0.2661269	0.1330327	0.5132997
flow_90:year2003	0.0094506	0.0001190	0.0092172	0.0096840	0.0000000
flow_90:year2004	-0.0101387	0.0002436	-0.0106163	-0.0096610	0.0000000
flow_90:year2005	-0.0046105	0.0005766	-0.0057411	-0.0034799	0.0000000

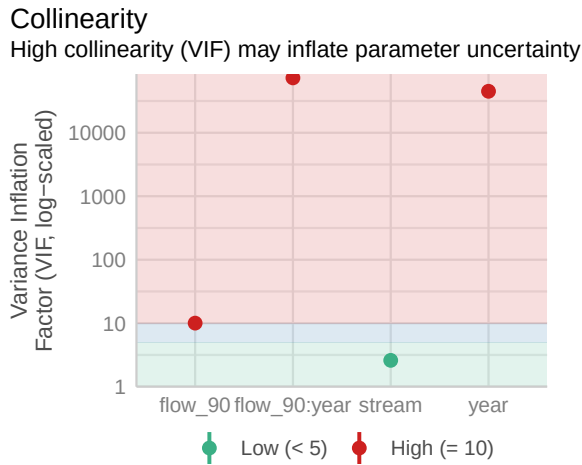


Figure A2.9: Variance inflation factors for model m.flow.7

AIC scores suggest `m.flow.7` is best model (Table A2.9). However, while the R^2 value is 0.98, the parameter estimate for `flow_90` has is incredibly small and most variation is now attributed to the year contrasts (Table A2.10). Thus, `flow_90` is clearly confounded with `year`, and confirmed by high Variance Inflation Factor (VIF) scores (Figure A2.9).

A2.5.1 Why `flow_90` is problematic

Not spatially resolved

- We are modeling spawn timing at the redd level (COMID/stream)
- But `flow_90` is calculated from a single downstream gauge, and applied to all redds
- This assumes flow conditions are identical across all sites
- Including it gives the illusion of spatially resolved variation that isn't there

Correlated with year

- Since `flow_90` varies mostly across years, it is strongly confounded with year
- Any flow-related signal is probably already captured by your year fixed effect
- Including both `flow_90` and year risks collinearity, and may produce misleading inferences

Spawn-time aligned flow does not equal experienced flow

- While `flow_90` is aligned to each redd's spawn date, it still reflects a lower-basin gauge, not the actual hydrologic conditions experienced at the redd site
- So it might be precisely wrong — aligned in time but irrelevant in space

Recommendation: Drop `flow_90` from model.

Although we initially considered including 90-day mean streamflow (`flow_90`) as a predictor of spawn timing, this variable was ultimately excluded due to concerns about ecological validity and model overfitting. Stream flow data were derived from a single downstream USGS gauge and did not capture spatial variation across the study streams or reaches. Moreover, because `flow_90` was closely aligned with year, it introduced strong collinearity with the year effect and risked attributing site-level variation to flow patterns not actually experienced by individual redds. As such, we excluded `flow_90` to avoid misleading inference.

A3 Data Analysis

The final analysis dataset includes:

- `yday`: spawn date, continuous response variable
- `comid`, `stream`, `year`: grouping variables
- `temp_90`: 90-day mean temperature pre-spawn, continuous predictor variable
- `slope` and `mean_elevation`: provisionally retaining continuous predictor variable

We used linear mixed-effects models to evaluate environmental predictors of Chinook salmon spawn timing, with redd observation day-of-year (`yday`) as the response variable. We scaled the continuous covariates to have a mean of 0 and standard deviation of 1 to assist convergence and interpretation. `COMID` will be included in all models as a random (intercept) effect to account for repeated measures on `COMIDs`. `Stream` will be treated as a fixed effect to compare average effects across streams, and `year` as a fixed effect to compare average effects across years. Further, based on exploratory analysis and biological expectations of nonlinear thermal responses, temperature effects were modeled using both linear and quadratic terms for the 90-day average stream temperature prior to spawning (`temp_90` and `I(temp_90^2)`).

A3.1 Initial Model Selection

We fit 31 additive models representing all combinations of fixed effects: `temp_90`, `stream`, `year`, `slope`, and `mean_elevation`. Model selection was performed using AIC, and model fit was evaluated using marginal and conditional R^2 , RMSE, and intraclass correlation coefficients (ICC).

```
# one covariate
m1 <- lmer(yday ~ temp_90 + I(temp_90^2) + (1 | COMID), data = df_mod,
  REML = FALSE)
m2 <- lmer(yday ~ stream + (1 | COMID), data = df_mod, REML = FALSE)
m3 <- lmer(yday ~ year + (1 | COMID), data = df_mod, REML = FALSE)
m4 <- lmer(yday ~ mean_elevation + (1 | COMID), data = df_mod, REML = FALSE)
m5 <- lmer(yday ~ slope + (1 | COMID), data = df_mod, REML = FALSE)

# two covariates
m6 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + (1 | COMID), data = df_mod,
  REML = FALSE)
m7 <- lmer(yday ~ temp_90 + I(temp_90^2) + year + (1 | COMID), data = df_mod,
  REML = FALSE)
m8 <- lmer(yday ~ temp_90 + I(temp_90^2) + mean_elevation + (1 | COMID),
  data = df_mod, REML = FALSE)
m9 <- lmer(yday ~ temp_90 + I(temp_90^2) + slope + (1 | COMID), data = df_mod,
  REML = FALSE)
m10 <- lmer(yday ~ stream + year + (1 | COMID), data = df_mod, REML = FALSE)
m11 <- lmer(yday ~ stream + mean_elevation + (1 | COMID), data = df_mod,
  REML = FALSE)
m12 <- lmer(yday ~ stream + slope + (1 | COMID), data = df_mod, REML = FALSE)
m13 <- lmer(yday ~ year + mean_elevation + (1 | COMID), data = df_mod,
  REML = FALSE)
m14 <- lmer(yday ~ year + slope + (1 | COMID), data = df_mod, REML = FALSE)
m15 <- lmer(yday ~ mean_elevation + slope + (1 | COMID), data = df_mod,
  REML = FALSE)

# three covariates
```

```

m16 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + year + (1 | COMID),
  data = df_mod, REML = FALSE)
m17 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + mean_elevation + (1 |
  COMID), data = df_mod, REML = FALSE)
m18 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + slope + (1 | COMID),
  data = df_mod, REML = FALSE)
m19 <- lmer(yday ~ temp_90 + I(temp_90^2) + year + mean_elevation + (1 |
  COMID), data = df_mod, REML = FALSE)
m20 <- lmer(yday ~ temp_90 + I(temp_90^2) + year + slope + (1 | COMID),
  data = df_mod, REML = FALSE)
m21 <- lmer(yday ~ temp_90 + I(temp_90^2) + mean_elevation + slope + (1 |
  COMID), data = df_mod, REML = FALSE)
m22 <- lmer(yday ~ stream + year + mean_elevation + (1 | COMID), data = df_mod,
  REML = FALSE)
m23 <- lmer(yday ~ stream + year + slope + (1 | COMID), data = df_mod,
  REML = FALSE)
m24 <- lmer(yday ~ stream + mean_elevation + slope + (1 | COMID), data = df_mod,
  REML = FALSE)
m25 <- lmer(yday ~ year + mean_elevation + slope + (1 | COMID), data = df_mod,
  REML = FALSE)

# four covariates
m26 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + year + mean_elevation +
  (1 | COMID), data = df_mod, REML = FALSE)
m27 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + year + slope + (1 |
  COMID), data = df_mod, REML = FALSE)
m28 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + mean_elevation + slope +
  (1 | COMID), data = df_mod, REML = FALSE)
m29 <- lmer(yday ~ temp_90 + I(temp_90^2) + year + mean_elevation + slope +
  (1 | COMID), data = df_mod, REML = FALSE)
m30 <- lmer(yday ~ stream + I(temp_90^2) + year + mean_elevation + slope +
  (1 | COMID), data = df_mod, REML = FALSE)

# all five covariates
m31 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + year + mean_elevation +
  slope + (1 | COMID), data = df_mod, REML = FALSE)

```

Table A3.1: AIC selection performance metrics for additive linear models; top 10 of 31 shown.

Model	df	AIC	delta_AIC	R2_marginal	R2_conditional	RMSE	ICC
m26	16	13356.04	0.000000	0.7852656	0.9560890	2.023039	0.7955103
m31	17	13358.04	1.997893	0.7852838	0.9560902	2.023041	0.7954984
m27	16	13468.96	112.924527	0.7366068	0.9794720	2.022170	0.9220632
m16	15	13473.12	117.079398	0.7293886	0.9800601	2.022117	0.9263154
m29	10	13487.82	131.782678	0.7272376	0.9840770	2.021709	0.9416233
m19	9	13488.73	132.689748	0.7256116	0.9843959	2.021715	0.9431312
m7	8	13531.17	175.127777	0.6600787	0.9871140	2.021778	0.9620912
m20	9	13532.90	176.858004	0.6602599	0.9870887	2.021778	0.9619966
m17	13	15787.25	2431.209763	0.7827740	0.8796472	3.112216	0.4459559
m28	14	15789.04	2433.003132	0.7834455	0.8797931	3.112327	0.4449113

Based on AIC, the best-fitting model was m26, though the improvement over m31 (the full model including

`slope`) was modest, suggesting little benefit to including `slope` as a predictor (Table A3.1).

`m27`, which excludes `mean_elevation`, had substantially worse AIC and performance than `m26` and `m31`, indicating that elevation contributes meaningfully to explaining spawn timing variation.

Model `m16`, which excludes both `mean_elevation` and `slope`, had nearly identical predictive accuracy (RMSE) and higher conditional R^2 compared to `m26` and `m31`, despite a slightly lower (0.73 vs 0.78) marginal R^2 . The 117-point AIC difference between `m16` and `m26` likely reflects subtle improvements in likelihood fit due to topographic covariates.

Visual inspection of partial (model-based) relationships (Figure A3.1 B)) revealed that the modeled elevation effect was inconsistent with ecological expectations.

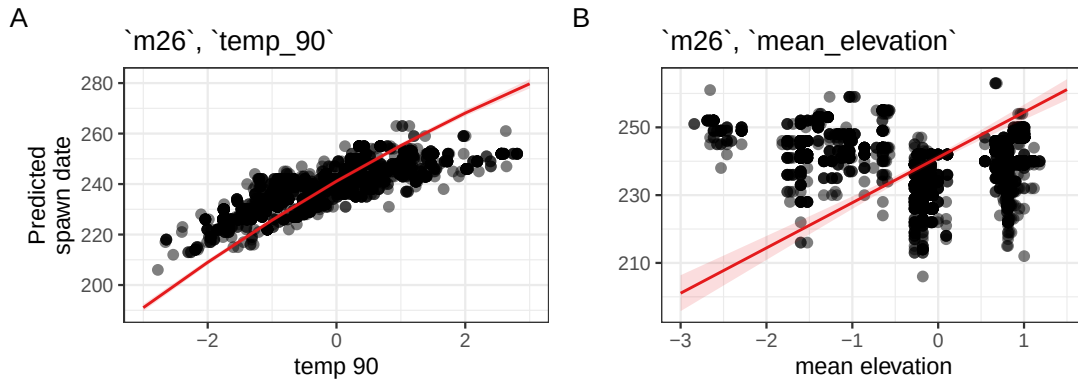


Figure A3.1: Predicted relationship (red line) between spawn date ('yday') and (A) 'temp_90', (B) 'mean_elevation'.

In this case, `m26` is picking up a positive association: after controlling for temperature and stream identity, higher-elevation reaches are predicted to spawn later. This is contrary to the observed negative relationships between elevation, temperature, and spawn timing across streams (Figure A2.6), and could reflect either a real ecological effect or a statistical artifact.

There are plausible ecological mechanisms for a residual positive elevation effect. Cooler conditions and delayed snowmelt at higher elevations could push back spawning timing independently of mean temperature. Higher-elevation reaches may also host subpopulations adapted to shorter growing seasons, or simply take longer for migrating fish to reach. These remain second-order hypotheses that would require stronger ecological justification to rely on.

Variance inflation factors (VIFs) from `m26` indicated moderate collinearity between elevation and stream (VIF 2248 6.5 for each). Elevation is strongly confounded with stream identity in this dataset (Figure A2.6), and the statistical case for caution is more compelling than the ecological one: once temperature is partialled out, the remaining variation in elevation is limited and potentially unstable, and inconsistent effects across streams suggest this is not a robust, generalizable signal.

Both interpretations are acknowledged: the positive elevation effect emerges statistically after controlling for temperature but is likely confounded by collinearity; and while a true ecological effect is possible, the available data cannot disentangle these mechanisms.

A3.2 Targeted model comparison

We compared our top additive model (`m16`) to a series of increasingly flexible models to evaluate the contribution of random slopes and temperature nonlinearity.

A3.2.1 Random slopes for temp_90

To account for potential variation in temperature sensitivity across sites, we extended `m16` to include **COMID-specific random slopes for temp_90**, yielding a model with the same fixed effects but with the random structure $(1 + \text{temp_90} \mid \text{COMID})$. Because the random effects structure differed, we fit both models using restricted maximum likelihood (REML).

```
m16_reml <- lmer(
  yday ~ temp_90 + I(temp_90^2) + stream + year + (1 | COMID),
  data = df_mod, REML = TRUE
)
m16_rs_reml <- lmer(
  yday ~ temp_90 + I(temp_90^2) + stream + year + (1 + temp_90 | COMID),
  data = df_mod, REML = TRUE
)
```

Table A3.2: Model selection for ‘m16’ vs. ‘m16_rs’.

Model	df	AIC	delta_AIC	R2_marginal	R2_conditional	RMSE	ICC
m16_rs_reml	17	12948.76	0.000	0.698	0.985	1.781	0.949
m16_reml	15	13459.05	510.283	0.714	0.981	2.022	0.932

The random slope model (`m16_rs_reml`) substantially improved fit ($\Delta\text{AIC} = 510$), reduced RMSE (2.02 $\rightarrow$ 1.78 days), and slightly increased conditional R^2 (0.981 $\rightarrow$ 0.985). However, the marginal R^2 declined from 0.714 to 0.698, reflecting a redistribution of explanatory power from fixed to random effects. This shift is expected when site-specific variation in temperature responses is modeled explicitly, and suggests that temperature sensitivity varies meaningfully among stream reaches. We next assessed residual behavior of `m16_rs_reml` to evaluate whether the quadratic temperature term adequately captured the nonlinear relationship before proceeding to functional form comparison.

A3.3 Residual Diagnostics

DHARMa diagnostics for `m16_rs_reml` revealed adequate overall fit alongside a systematic pattern that motivated evaluation of a more flexible temperature functional form. The QQ plot showed close tracking of the diagonal despite a flagged KS test ($p \approx 0$) and outlier test ($p = 1 \times 10^{-5}$); with $n = 3,016$, DHARMa’s omnibus tests readily detect minor deviations that carry little practical significance, and the dispersion test was non-significant ($p = 0.44$). More informative was the residuals vs. predicted plot, which showed a pronounced S-shaped pattern in the quantile smoothers — indicating mild non-uniformity across the prediction range, likely reflecting asymmetry in spawn timing at the distributional tails.

The residuals vs. `temp_90` plot was more diagnostic: quantile smoothers showed a clear monotonic decline across the temperature gradient, with residuals shifted consistently above 0.5 at cold sites and below 0.5 at warm sites. This systematic trend indicates the model was over-predicting spawn timing at low temperatures and under-predicting at high temperatures even after the quadratic term, suggesting the quadratic did not fully capture the nonlinear temperature–spawn timing relationship. Residuals plotted against stream and year were also flagged (Levene test significant), reflecting heterogeneity not fully absorbed by the fixed effects: Elk and Marsh creeks showed compressed distributions shifted below 0.5, while Big and Camas were shifted high, and 2005 exhibited notably lower residuals than other years.

Together, these patterns motivated evaluation of a more flexible functional form for the temperature effect.

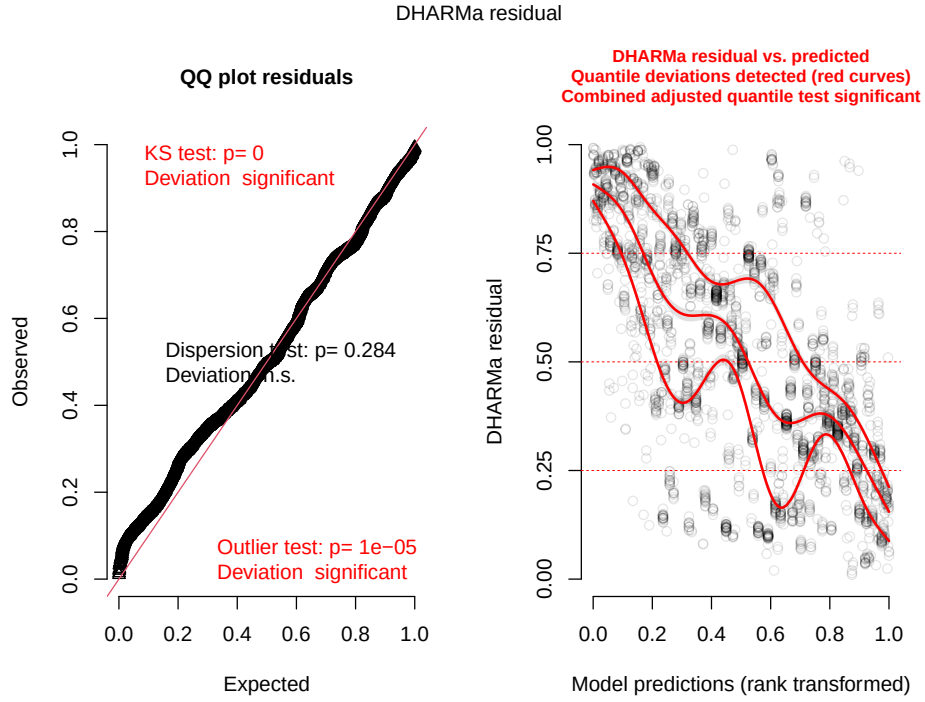


Figure A3.2: DHARMA residuals 1.

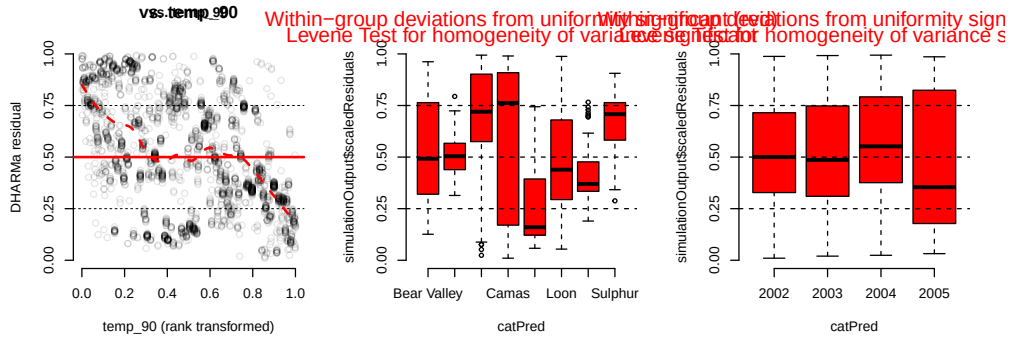


Figure A3.3: DHARMA residuals 2.

A3.4 Evaluation of Temperature Functional Form

The $df=3$ spline substantially outperformed the quadratic ($\Delta AIC = 139$) with lower RMSE and improved residual behavior across the temperature gradient. The $df=4$ spline showed further AIC improvement ($\Delta AIC = 66$ over $df=3$) but produced biologically implausible predictions at warm-end temperature extremes — spawn timing decreasing sharply above $\sim 13^{\circ}\text{C}$ — indicative of overfitting. The $df=3$ spline was therefore selected as the final temperature functional form.

Table A3.3: AIC comparison of quadratic vs. natural spline temperature terms.

Model	df	AIC	delta
mod_spline4	19	12761.28	0.00000
mod_spline3	18	12827.09	65.81274
m16_rs	17	12965.97	204.69238

Table A3.4: Model performance for quadratic vs. spline formulations.

Name	R2_conditional	R2_marginal	ICC	RMSE	Performance_Score
mod_spline4	0.9950744	0.4831279	0.9904704	1.666518	0.8333333
mod_spline3	0.9903689	0.6087398	0.9753845	1.715325	0.3942136
m16_rs	0.9840995	0.7132881	0.9445417	1.781828	0.1666667

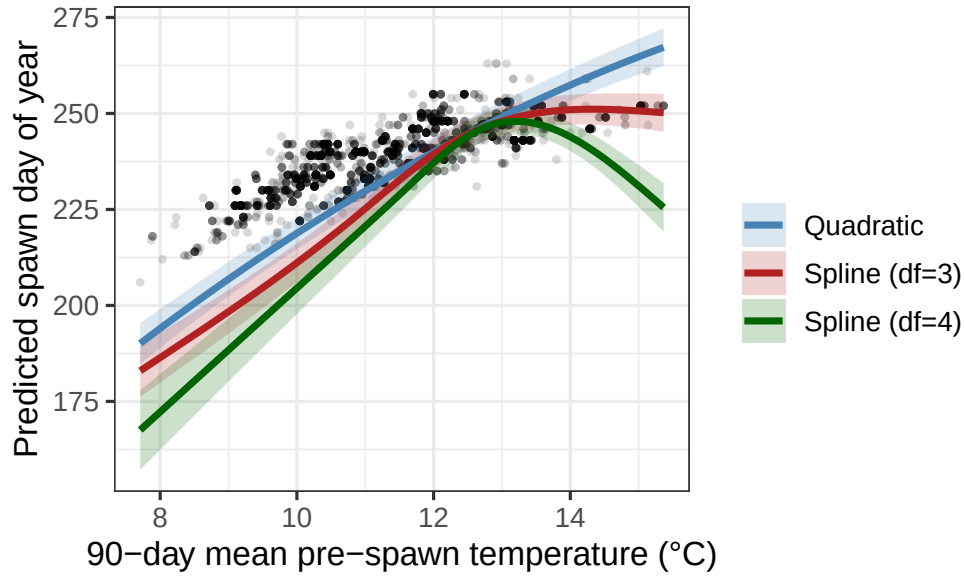
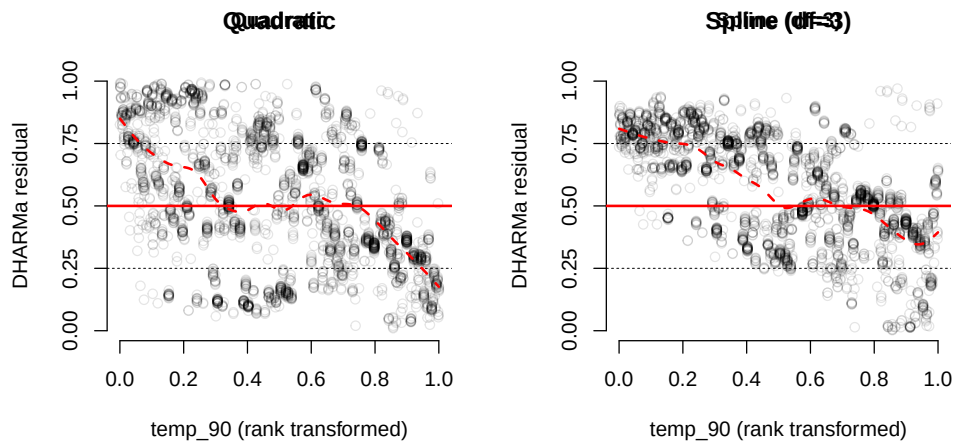


Figure A3.4: Prediction comparison plots.



A3.5 Interactions

To assess whether fixed-effect interactions provide additional explanatory power beyond what is captured by random slopes, we tested two models: one with a `ns(temp_90, df=3) %times% stream` interaction (`m201`) and one with a `ns(temp_90, df=3) %times% year` interaction (`m202`). Both were compared to the spline baseline (`mod_spline3`).

```
m201 <- lmer(
  yday ~ ns(temp_90, df = 3) * stream + year + (1 + temp_90 | COMID),
  data = df_mod, REML = FALSE)
m202 <- lmer(
  yday ~ ns(temp_90, df = 3) * year + stream + (1 + temp_90 | COMID),
  data = df_mod, REML = FALSE)
```

Table A3.5: Model selection for interaction models vs. ‘`mod_spline3`’.

Model	df	AIC	delta_AIC	R2_marginal	R2_conditional	RMSE	ICC
m202	27	11345.38	0.000	0.6526270	0.9957698	1.297116	0.9878223
m201	39	12551.32	1205.931	0.6288294	0.9916978	1.594317	0.9776324
mod_spline3	18	12827.09	1481.705	0.6087398	0.9903689	1.715325	0.9753845

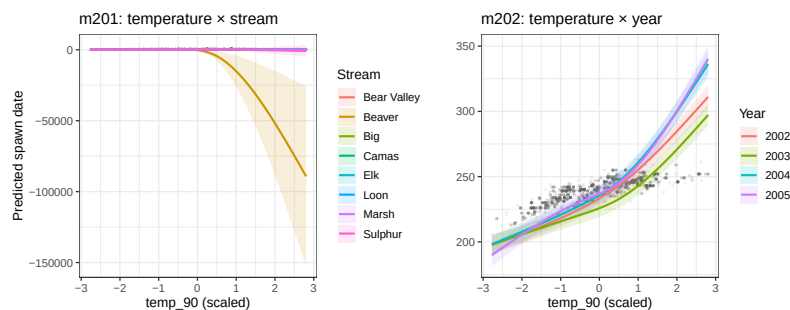


Figure A3.5: Predicted spawn timing for interaction models.

Model `m201` (temperature $\times$ stream interaction) produced severely implausible predictions: the Beaver Creek-specific spline diverged catastrophically outside its observed temperature range, with predicted spawn timing declining to biologically impossible values. This behavior reflects data sparsity — Beaver Creek redds span a narrow temperature range, leaving the interaction term unconstrained at extrapolation, a well-known hazard of spline interactions with unbalanced groups. Although predictions for other streams appeared reasonable, the instability of a single group’s interaction term is sufficient grounds for exclusion.

Model `m202` (temperature $\times$ year interaction) produced biologically plausible year-specific curves and a large AIC improvement ($\Delta\text{AIC} = 1,482$ over `mod_spline3`). However, this improvement likely reflects the interaction absorbing interannual differences in the temperature distribution rather than genuine heterogeneity in the temperature–spawn timing relationship across years — a form of overfitting that capitalizes on year-specific data ranges rather than estimating a stable biological effect. Consistent with this interpretation, variance explained (R^2) and RMSE improved modestly relative to the large reduction in AIC.

Together, these results confirm that interaction terms introduce instability or overfit without providing reliable biological inference. We therefore retained `mod_spline3` as the final model, which captures population-level nonlinearity in the temperature response while allowing reach-specific deviations through random slopes.

A4 Final Model Diagnostics and Interpretation

```
# Final model: natural spline (df=3) for temperature, COMID random slopes, fit with REML
mod_final <- lmer(
  yday ~ ns(temp_90, df = 3) + stream + year + (1 + temp_90 | COMID),
  data = df_mod, REML = TRUE
)
```

A4.1 Residual diagnostics

Diagnostics for the final model (`mod_final`) indicated strong overall performance alongside several patterns that warrant transparent reporting (Figures A4.1, A4.2, A4.3).

The posterior predictive check showed excellent agreement between observed and model-predicted distributions of spawn timing, with nearly overlapping density curves and no systematic deviation in the central tendency or spread. Collinearity was low, with variance inflation factors (VIFs) for all fixed effects well below 5. Influential observations were present — several redds showed moderate-to-high leverage — but standardized residuals remained within acceptable bounds and no observations were extreme enough to justify removal.

The residual vs. fitted and homogeneity of variance plots showed mild heteroscedasticity, with slight funneling at lower fitted values. This pattern likely reflects the skewed distribution of early spawn dates rather than model misspecification, and was insufficient to invalidate inference. The normality of residuals Q-Q plot showed more pronounced tail deviations — particularly in the upper tail — suggesting the Gaussian error assumption is imperfect. Random effect Q-Q plots for both intercepts and slopes tracked the diagonal reasonably well with modest tail departures, indicating acceptable distributional assumptions for the mixed-effects structure.

DHARMA simulated residuals corroborated these patterns while providing additional diagnostic nuance. The KS test ($p \approx 0$), dispersion test ($p = 0.004$), and outlier test ($p = 1 \times 10^{-5}$) were all flagged significant; however, with $n = 3,016$, DHARMA’s omnibus tests are highly sensitive and will reliably detect minor deviations that carry little inferential consequence. More informative were the directional patterns: the residuals vs. predicted plot showed a residual S-shaped pattern in the quantile smoothers, and residuals plotted against stream and year remained flagged, indicating that the fixed effects for these grouping variables do not fully account for distributional heterogeneity across sites and years. Importantly, the residuals vs. `temp_90` plot showed a substantially flatter quantile trend compared to the quadratic model, confirming that the natural spline captured the nonlinear temperature–spawn timing relationship more faithfully.

Taken together, these diagnostics indicate that the final model is well-specified for its primary inferential purpose — characterizing the population-level temperature–spawn timing relationship and its spatial heterogeneity — while acknowledging that the Gaussian error structure imperfectly represents the distributional properties of spawn dates. Crucially, the persistent residual structure across streams and years is not a modeling failure but an ecological signal: it reflects fine-scale habitat heterogeneity — groundwater connectivity, hyporheic exchange, local geomorphology, run composition — that operates below the resolution of available covariates and cannot be absorbed by stream identity or year alone. This is precisely the unmeasured variation that the random slope structure was designed to partially accommodate, and its persistence in the residuals reinforces the central finding of this paper: that spawn timing diversity in the Middle Fork is sustained by reach-scale processes that temperature measurements and fixed grouping variables, however carefully assembled, cannot fully characterize.

The reiterate: the story is not “the model doesn’t fit perfectly”, the story is “the residuals are showing you the thing the paper is about.” That’s a genuinely defensible position given the ICC and posterior predictive check, and it connects this diagnostic section directly to the ecological argument rather than leaving it as a standalone technical caveat.

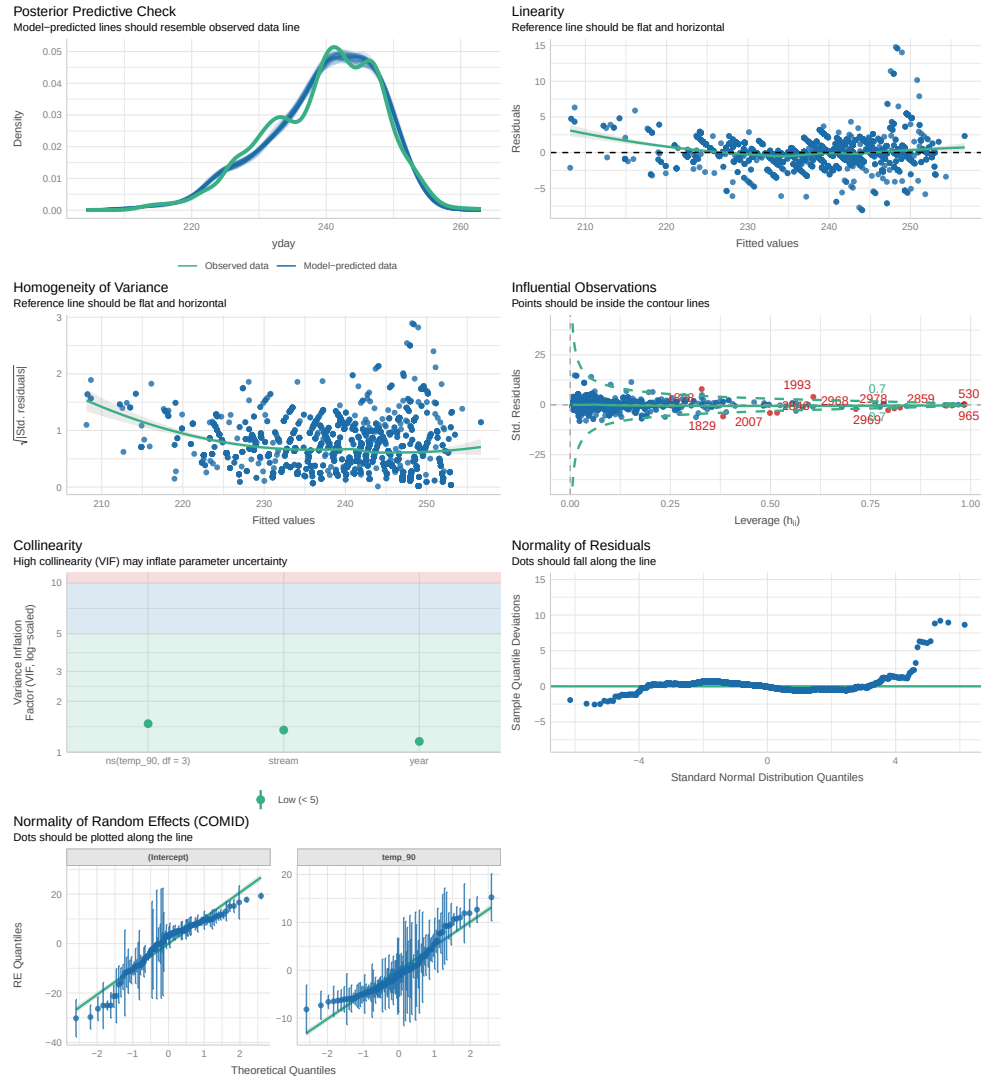


Figure A4.1: Residual diagnostics for mod_final.

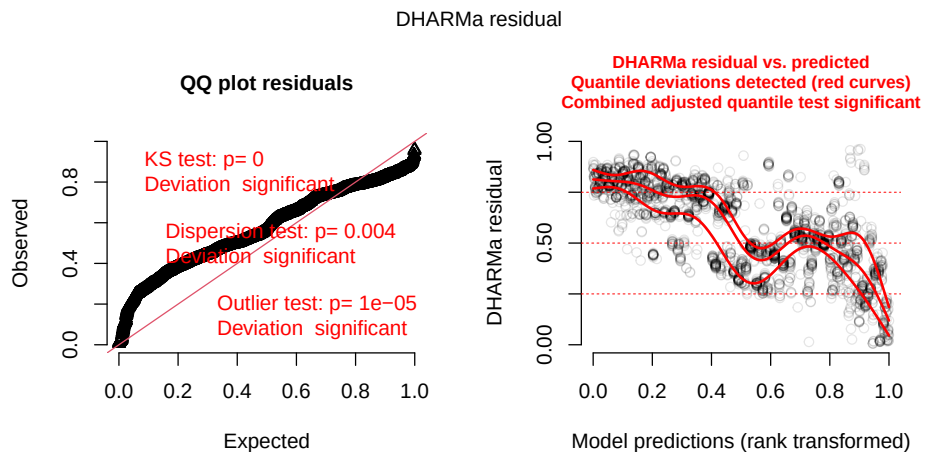


Figure A4.2: DHARMa residuals 1.

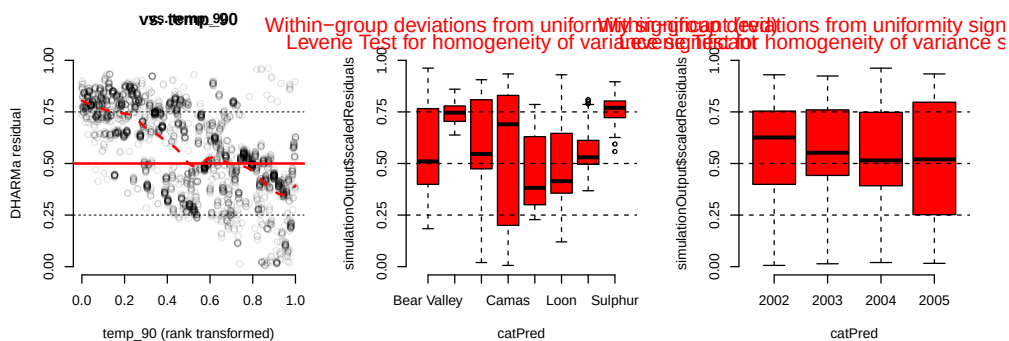


Figure A4.3: DHARMa residuals 2.

A4.2 Model parameters and performance

Parameter estimates (Table 2) confirm a significant nonlinear effect of 90-day antecedent temperature on spawn timing. The three natural spline basis terms were each highly significant (all $p < 0.001$), indicating that the temperature–spawn timing relationship departs meaningfully from linearity across the observed thermal gradient; the shape of this relationship is best interpreted visually (Figure 5) rather than from the basis coefficients, which have no direct biological interpretation individually.

Most stream and year effects were significant, capturing expected spatiotemporal structure in spawn timing (Table 2). Beaver, Elk, and Sulphur creeks showed the largest positive deviations from the Bear Valley reference (estimates of 14.88, 12.23, and 13.44 days respectively), while Big, Loon, and Marsh creeks were not significantly different from Bear Valley — consistent with their overlapping thermal regimes and elevations. Year effects were all significant, with 2003 associated with earlier spawning (-4.83 days) and 2004–2005 with progressively later spawning, reflecting interannual variability in thermal and flow conditions.

The random effects structure reveals the dominant source of variation in spawn timing. The intercept variance ($\tau_{00} = 128.20$) and slope variance ($\tau_{11} = 33.44$) indicate substantial among-COMID heterogeneity in both baseline timing and thermal sensitivity — reach-level deviations are large relative to the fixed effects, confirming that local habitat conditions shape spawn timing in ways the measured covariates cannot fully characterize. The strong negative intercept-slope correlation ($\rho_{01} = -0.83$) indicates that reaches with later baseline spawn timing exhibit substantially weaker thermal sensitivity, while earlier-spawning reaches show stronger temperature dependence — a pattern consistent with early-spawning reaches occupying more thermally dynamic, higher-elevation positions where temperature is a more variable and reliable phenological cue, while later-spawning, lower-elevation reaches may be approaching physiological or environmental constraints that buffer their thermal response. The ICC of 0.98 formalizes this: knowing which reach a redd belongs to accounts for 98% of total variance in spawn timing, with the remaining 2% distributed as residual error. As discussed in the diagnostics section, the residual structure within that 2% carries a genuine ecological signal — the fine-scale hydrologic and geomorphic processes that produce characteristic reach-level timing signatures — rather than reflecting model inadequacy.

Model fit was strong: marginal R^2 (fixed effects only) was 0.599 and conditional R^2 (fixed + random effects) was 0.990 (Table 2). The gap between these values reflects the predominance of the random effects — particularly the slopes — in explaining spawn timing variation, with fixed effects providing the population-level scaffold on which reach-specific responses are structured. Notably, the marginal R^2 of the spline model (0.599) is modestly lower than the quadratic (0.713), reflecting the spline’s redistribution of explanatory power toward the random effects rather than a loss of predictive performance — conditional R^2 and RMSE both improved.

Table A4.1: Model selection for linear mixed-effects models of Chinook salmon spawn timing ($n = 3,016$ redds; 104 COMIDs; 8 streams; 2002–2005). Models were evaluated sequentially across four decision points: fixed-effect structure, random slope inclusion, temperature functional form, and interaction terms. ΔAIC is computed relative to the reference model (bold) within each group; negative values indicate better AIC than the reference. RMSE in days. R^2 marginal = variance explained by fixed effects only; R^2 conditional = fixed + random effects combined. Models marked † were excluded despite lower AIC due to biologically implausible predictions or overfitting (see Appendix A3).

Model	ΔAIC	RMSE	R^2 (marg.)	R^2 (cond.)	Notes
1. Additive models (random intercept only)					
m16: temp_90 + stream + year	0.0	2.02	0.729	0.980	Selected base model
m26: m16 + elevation†	-117.1	2.02	0.785	0.956	Collinearity and signal captured by random structure; excluded
m31: m26 + slope†	-115.1	2.02	0.785	0.956	
2. Random slope evaluation (ML fit)					
m16_rs: + random slopes for temp_90	0.0	1.78	0.713	0.984	$\Delta\text{AIC} = 507.1$ vs. intercept-only
m16: random intercept only	507.1	2.02	0.729	0.980	
3. Temperature functional form (ML fit)					
Natural spline df = 3	0.0	1.72	0.609	0.990	Selected final functional form
Quadratic: temp_90 + temp_90 ²	138.9	1.78	0.713	0.984	
Natural spline df = 4†	-65.8	1.67	0.483	0.995	Biologically implausible at high temperatures; excluded
4. Interaction terms (ML fit)					
Final model: no interactions	0.0	1.72	0.609	0.990	Spline extrapolation unstable in data-sparse streams; excluded
+ temperature × stream	-275.8	1.59	0.629	0.992	
+ temperature × year†	-1481.7	1.30	0.653	0.996	Large AIC gain likely reflects year-specific data ranges, not biological interaction; excluded

Table A4.2: Parameter estimates for the final linear mixed-effects model of Chinook salmon spawn timing: $y_{\text{day}} \sim \text{ns}(\text{temp_90}, \text{df} = 3) + \text{stream} + \text{year} + (1 + \text{temp_90} \mid \text{COMID})$, fit by REML. Fixed effects are shown with estimates, 95% confidence intervals, and p -values. Reference levels: Stream = Bear Valley Creek; Year = 2002. temp_90 was scaled to mean = 0, SD = 1 prior to fitting. Natural spline basis terms (df = 3) are not individually interpretable; see Figure 5 for the estimated temperature–spawn timing relationship. Random effects summarise among-reach variance in baseline timing (τ_{00}), thermal sensitivity (τ_{11}), and their correlation (ρ_{01}). ICC = intraclass correlation coefficient.

Parameter	Est.	95% CI	p
Fixed effects			
Intercept	183.11	[176.16, 190.05]	<0.001
Temp. spline term 1	59.61	[55.29, 63.93]	<0.001
Temp. spline term 2	94.54	[85.83, 103.25]	<0.001
Temp. spline term 3	52.14	[46.22, 58.06]	<0.001
Stream: Beaver	14.88	[7.75, 22.02]	<0.001
Stream: Big	3.84	[-1.11, 8.80]	0.128
Stream: Camas	5.53	[0.28, 10.77]	0.039
Stream: Elk	12.23	[5.08, 19.37]	<0.001
Stream: Loon	4.15	[-0.86, 9.16]	0.104
Stream: Marsh	5.25	[-0.34, 10.84]	0.065
Stream: Sulphur	13.44	[7.02, 19.85]	<0.001
Year: 2003	-4.83	[-5.05, -4.60]	<0.001
Year: 2004	2.76	[2.53, 2.99]	<0.001
Year: 2005	3.72	[3.42, 4.01]	<0.001
Random effects			
σ^2 (residual)	3.12	NA	NA
τ_{00} COMID (intercept)	128.20	NA	NA
τ_{11} COMID (slope)	33.44	NA	NA
ρ_{01} COMID (int. $\times$ slope)	-0.83	NA	NA
ICC	0.98	NA	NA
N (COMID)	104.00	NA	NA
N (obs.)	3016.00	NA	NA
Model performance			
R^2 marginal	0.60	NA	NA
R^2 conditional	0.99	NA	NA
RMSE (days)	1.71	NA	NA

A4.3 Predictions against original data

Model predictions closely matched observed spawn timing, with predicted values aligning well along the 1:1 line (Figure A4.4), consistent with the strong conditional R^2 reported above.

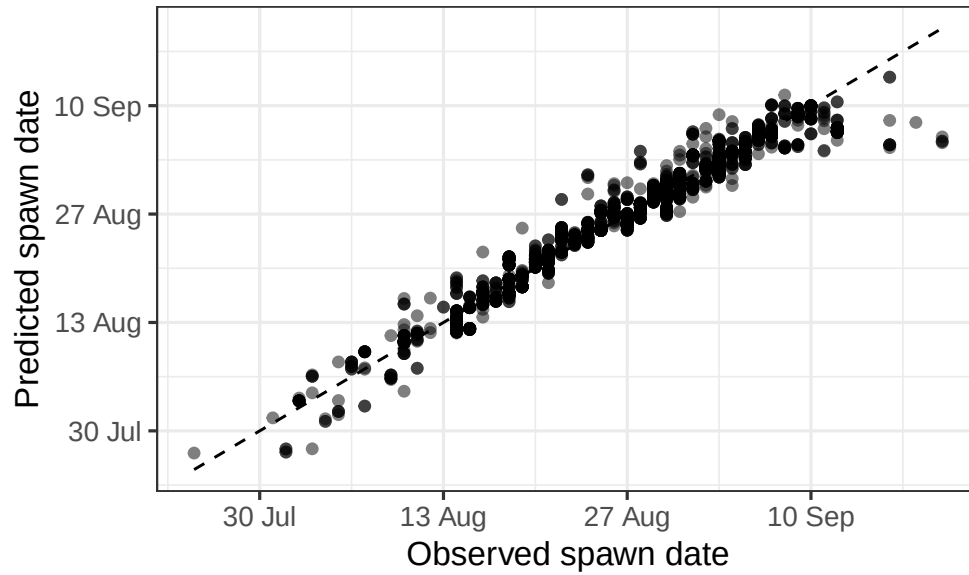


Figure A4.4: Predicted vs. observed spawn timing for Chinook salmon across all years and streams. The dashed line shows the 1:1 line. Points represent individual redd observations.

A4.4 Population-level effects

Marginal mean spawn timing was estimated at each factor level, averaging over random effects to characterize population-level patterns (Figure A4.5).

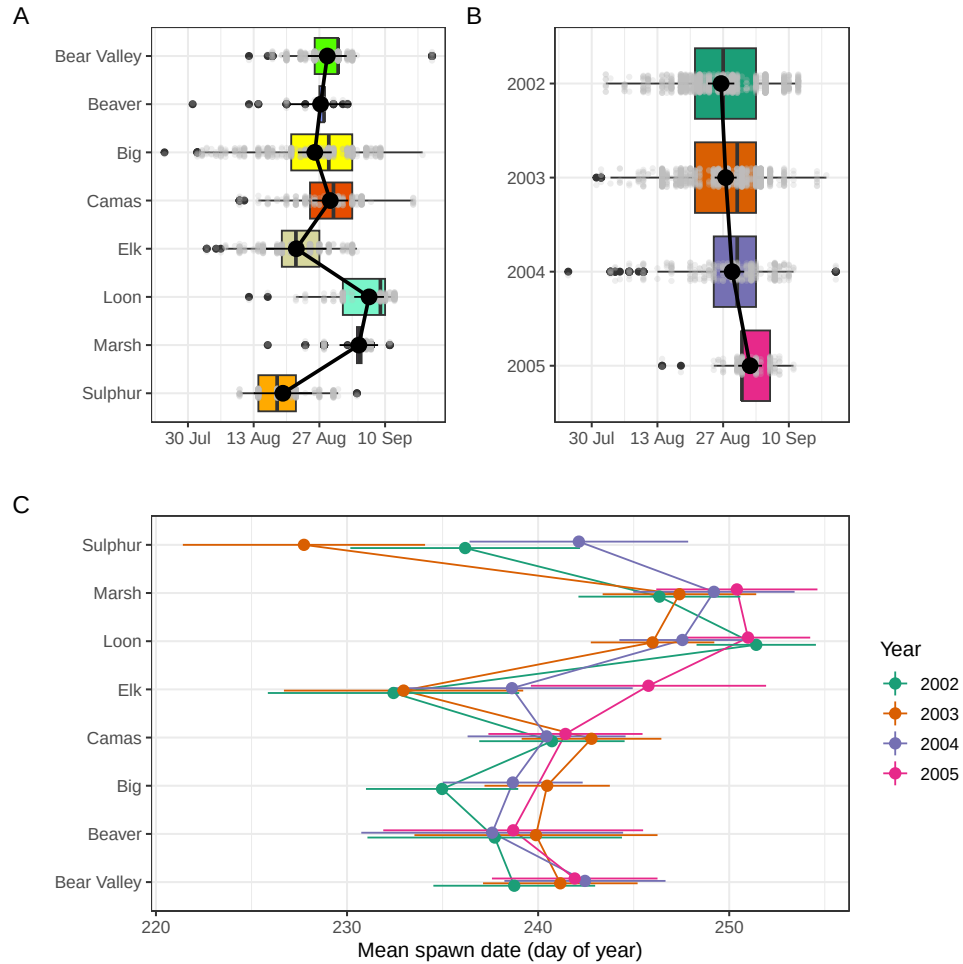


Figure A4.5: Predicted mean spawn dates by stream (A), year (B) and stream and year (C), from the final mixed-effects model. Black points with black lines (A, B), and colored points with horizontal lines (C) represent estimated marginal means and 95% confidence intervals. Boxplots in panels A and B show the distribution of observed redd counts by group, with individual data points in grey.

Significant differences in spawn timing were observed between many stream pairs (Table A4.3), particularly involving Loon (later spawning) and Sulphur (earlier spawning). For example, fish in Loon spawned significantly later than in Bear Valley, Camas, and Elk, while Sulphur exhibited significantly earlier timing than all other streams except Elk. These patterns reflect spatial heterogeneity in temperature and elevation across streams that is not fully captured by fixed effects alone.

Table A4.3: Pairwise contrasts among stream effects on predicted spawn timing. Contrasts represent estimated differences in predicted spawn day of year between streams from the final model. Positive values indicate later predicted spawn timing in the first stream compared to the second. P-values are uncorrected.

Level1	Level2	Difference	SE	CI_low	CI_high	t	df	p
Beaver	Bear Valley	-1.43	3.68	-8.65	5.80	-0.39	2998	0.70
Big	Bear Valley	-2.66	2.54	-7.65	2.33	-1.05	2998	0.30
Camas	Bear Valley	0.63	2.68	-4.62	5.88	0.24	2998	0.81
Elk	Bear Valley	-6.60	3.68	-13.83	0.62	-1.79	2998	0.07
Loon	Bear Valley	8.90	2.57	3.86	13.94	3.46	2998	0.00
Marsh	Bear Valley	6.72	2.86	1.11	12.32	2.35	2998	0.02
Sulphur	Bear Valley	-9.47	3.51	-16.35	-2.59	-2.70	2998	0.01
Big	Beaver	-1.23	3.37	-7.85	5.38	-0.37	2998	0.72
Camas	Beaver	2.06	3.50	-4.80	8.92	0.59	2998	0.56
Elk	Beaver	-5.18	4.17	-13.35	3.00	-1.24	2998	0.21
Loon	Beaver	10.33	3.57	3.33	17.32	2.89	2998	0.00
Marsh	Beaver	8.14	3.75	0.78	15.50	2.17	2998	0.03
Sulphur	Beaver	-8.04	3.80	-15.49	-0.60	-2.12	2998	0.03
Camas	Big	3.29	2.34	-1.31	7.89	1.40	2998	0.16
Elk	Big	-3.94	3.39	-10.59	2.70	-1.16	2998	0.24
Loon	Big	11.56	2.30	7.05	16.07	5.02	2998	0.00
Marsh	Big	9.38	2.60	4.27	14.48	3.60	2998	0.00
Sulphur	Big	-6.81	3.08	-12.86	-0.76	-2.21	2998	0.03
Elk	Camas	-7.23	3.52	-14.13	-0.34	-2.06	2998	0.04
Loon	Camas	8.27	2.43	3.51	13.03	3.41	2998	0.00
Marsh	Camas	6.09	2.72	0.75	11.43	2.23	2998	0.03
Sulphur	Camas	-10.10	3.27	-16.50	-3.70	-3.09	2998	0.00
Loon	Elk	15.50	3.56	8.53	22.47	4.36	2998	0.00
Marsh	Elk	13.32	3.75	5.97	20.67	3.55	2998	0.00
Sulphur	Elk	-2.87	3.88	-10.48	4.75	-0.74	2998	0.46
Marsh	Loon	-2.18	2.58	-7.23	2.87	-0.85	2998	0.40
Sulphur	Loon	-18.37	3.41	-25.06	-11.68	-5.39	2998	0.00
Sulphur	Marsh	-16.19	3.60	-23.24	-9.13	-4.50	2998	0.00

There was a clear trend toward later spawning over the four-year period (Table A4.4). Spawning in 2005 occurred significantly later than in all previous years. Differences between 2002 and 2003 were not statistically significant, but later years (2004 and especially 2005) were associated with a progressive delay in mean spawn timing. This temporal shift likely reflects interannual variability in temperature and flow conditions.

Table A4.4: Pairwise contrasts among year effects on predicted spawn timing. Contrasts represent estimated differences in predicted spawn day of year between years from the final model. Positive values indicate later spawning in the first year compared to the second. P-values are uncorrected.

Level1	Level2	Difference	SE	CI_low	CI_high	t	df	p
2003	2002	0.97	0.61	-0.23	2.18	1.59	2998	0.11
2004	2002	2.35	0.67	1.03	3.67	3.48	2998	0.00
2005	2002	6.18	0.64	4.92	7.43	9.64	2998	0.00
2004	2003	1.37	0.66	0.08	2.67	2.08	2998	0.04
2005	2003	5.20	0.75	3.74	6.66	6.98	2998	0.00
2005	2004	3.83	0.84	2.17	5.48	4.53	2998	0.00

A4.4.1 Predicted temperature–spawn timing relationships

Predicted spawn timing increased nonlinearly with 90-day mean pre-spawn temperature across all streams and years, leveling off at high temperatures consistent with biological expectations (Figure A4.6). Stream-to-stream variation reflects both fixed stream effects and COMID-specific random intercepts and slopes. Year-specific offsets were modest, with curvature consistent across years.

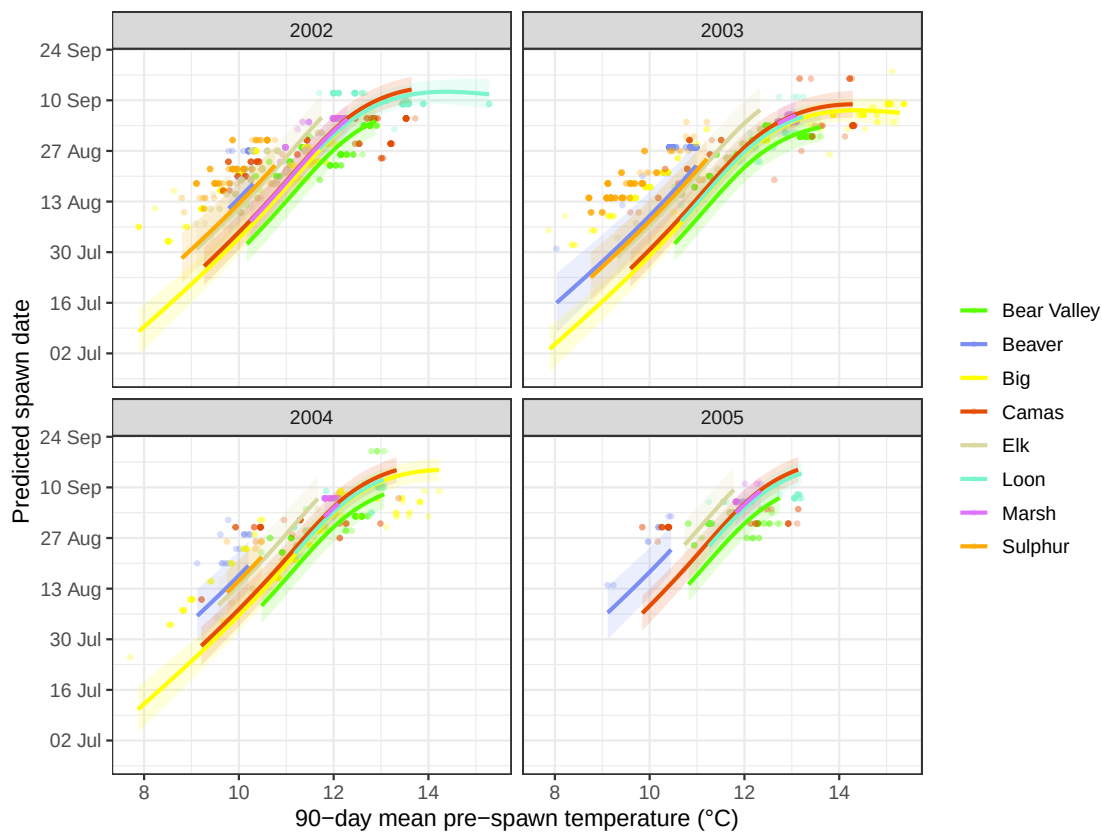


Figure A4.6: Predicted relationship between spawn timing and 90-day pre-spawn mean temperature by stream and year. Lines represent model predictions from the final mixed-effects model. Colored points show observed redd timing; shaded ribbons represent 95% confidence intervals.

A4.5 Group-level effects (deviations from fixed effects)

Random intercepts and slopes varied considerably among COMIDs, with substantial spread both within and between streams (Figure A4.7A, A4.7B). Random intercepts represent reach-level deviations in baseline spawn timing after accounting for fixed effects: a positive intercept indicates a reach where salmon spawn later than expected given temperature, stream identity, and year; a negative intercept indicates earlier-than-expected spawning. Random slopes represent deviations in thermal sensitivity: steeper slopes indicate stronger temperature dependence, shallower slopes reflect weaker or buffered responses. Among-reach variance was substantial for both baseline timing ($\sigma^2 = 128.20$) and thermal sensitivity ($\sigma^2 = 33.44$), and the pattern of variation differed among streams.

The intercept-slope correlation was strongly negative ($r = -0.83$; Figure A4.7C), indicating that reaches with earlier baseline spawn timing exhibited substantially greater thermal sensitivity. Early-spawning, likely higher-elevation reaches are not only earlier — they are also more responsive to temperature variation, suggesting greater phenotypic plasticity. In contrast, later-spawning, lower-elevation reaches exhibit shallower thermal responses, possibly reflecting operation near thermal thresholds or stronger constraint by other cues such as photoperiod or migration timing.

Figure A4.7D translates these random effect patterns into predicted temperature–spawn timing relationships for individual COMIDs, illustrating how the combination of varying intercepts and slopes generates a diverse family of reach-specific trajectories around the population mean. While the overall relationship between pre-spawn temperature and spawn timing is positive and nonlinear, the strength and shape of this response varies substantially among reaches, likely reflecting fine-scale differences in hydrology, geomorphology, and local thermal buffering capacity. This reach-level heterogeneity is the mechanism by which fine-scale habitat diversity sustains diverse spawning portfolios within the Middle Fork.

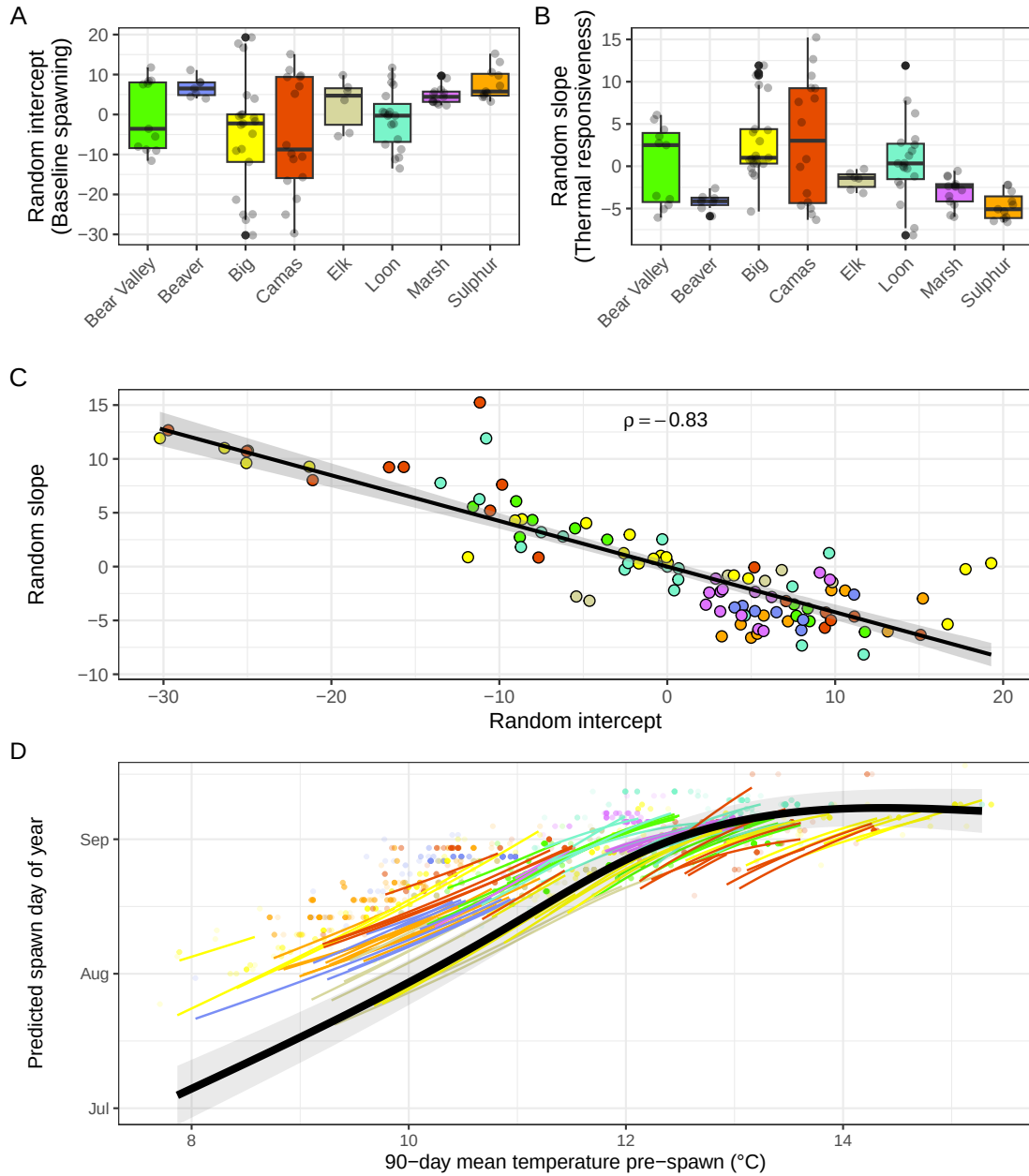


Figure A4.7: Reach-level (COMID) variation in random intercepts, random slopes, and predicted spawn timing relationships from the final mixed-effects model. (A) Boxplots of random intercepts by stream, representing reach-level deviations in baseline spawn timing after accounting for fixed effects (temperature, stream identity, and year); positive values indicate later-than-expected spawning, negative values indicate earlier-than-expected spawning. (B) Boxplots of random slopes by stream, representing reach-level deviations in thermal sensitivity (responsiveness of spawn timing to 90-day antecedent temperature); larger positive values indicate stronger temperature dependence. Points in A and B represent individual stream reaches (COMIDs). (C) Correlation between random intercepts and random slopes across all COMIDs, colored by stream. The strong negative relationship ($\rho = -0.83$) indicates that reaches with earlier baseline spawn timing exhibit substantially greater thermal sensitivity. The line represents a linear fit with 95% confidence interval. (D) Predicted temperature-spawn timing relationships for individual COMIDs (colored lines, colored by stream) and the population mean response (black line with 95% confidence interval). The diversity of reach-specific trajectories around the population mean reflects fine-scale variation in local habitat conditions — including hydrology, geomorphology, and thermal buffering — that structures phenological diversity across the Middle Fork watershed.